

CHAPTER ONE

NATURE OF OPERATIONS MANAGEMENT

“No great civilization has developed in isolation.”

Thomas Sewell

“Study the past if you would divine the future”

Confucius (490 B.C.)

Operations management has been recognized as an important factor in a country's economic growth. The traditional view of manufacturing management is the concept of **Production Management** with the focus on economic efficiency in manufacturing. Later the new name **Operations Management** was identified, as service sector became more prominent. Rapid changes in technology have posed numerous opportunities and challenges, which have resulted in enhancement of manufacturing capabilities through new materials, facilities, techniques and procedures. Hence, managing a service system has become a major challenge in the global competitive environment.

Operations Management has been a key element in the improvement and productivity in business around the world. Operations Management leads the way for the organizations to achieve its goals with minimum effort.

1.1 INTRODUCTION

Operation is as part of an organization, which is concerned with the transformation of a range of inputs into the required output (services) having the requisite quality level. Management is the process, which combines and transforms various resources used in the operations subsystem of the organization into value added services in a controlled manner as per the policies of the organization.

The set of interrelated management activities, which are involved in manufacturing certain products, is called as **production management**. If the same concept is extended to services management, then the corresponding set of management activities is called as **operations management**.

DEFINITIONS

There is no one single definition given to the term operations management (OM). The following are few of the definitions given by different writers.

- ❖ Operation management deals with the production of goods and services that people buy and use every day. It is a function that enables organization to achieve their goals through efficient acquisitions and utilization of resources.
- ❖ OM refers to the interaction and control of the process that transform input into finished goods and services.
- ❖ OM may be defined as the design, operation and improvement of the production systems that create the firm's primary products or services.
- ❖ OM may be defined as the management of the direct resources required to produce the goods and services provided by an organization. It is the derivative of the organization strategy and mission.

*Joseph G .Monks defines **Operations Management** as the process whereby resources, flowing within a defined system, are combined and transformed in a controlled manner to add value in accordance with policies communicated by management.*

The operations managers have the prime responsibility for processing inputs into outputs. They must bring together under production plan that effectively uses the materials, capacity and knowledge available in the production facility. Given a demand on the system work must be scheduled and controlled to produce goods and/or services required. Control must be exercised over such parameters such as costs, quality and inventory levels.

The definition of the operations management contains following **keywords:** *Resources, Systems, transformation and Value addition Activities.*

Resources

Resources are the human, material and capital inputs to the production process. Human resources are the key assets of an organization. As the technology advances, a large proportion of human input is in planning and controlling activities. By using the intellectual capabilities of people, managers can multiply the value of their employees into by many times. Material resources are the physical facilities and materials such as plant equipment, inventories and supplies. These are the major assets of an organization. Capital in the form of stock, bonds, and/or taxes and contributions is a vital asset. Capital is a store of value, which is used to regulate the flow of the other resources.

Systems

Systems are the arrangement of components designed to achieve objectives according to the plan. The business systems are subsystem of large social systems. In turn, it contains subsystem such as personnel, engineering, finance and operations, which will function for the good of the organization. A systems approach to operations management recognizes the hierarchical management responsibilities. If subsystems goals are pursued independently, it will results in sub-optimization. A consistent and integrative approach will lead to optimization of overall system goals.

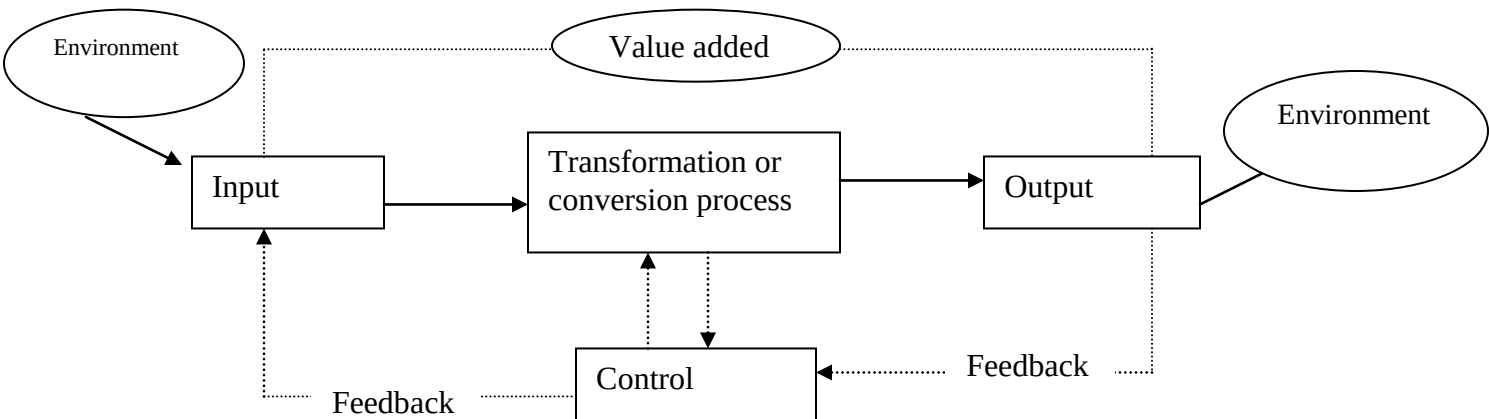
The ability of any system to achieve its objective depends on its design and its control. System design is a predetermined arrangement of components. It establishes the relationships that must exist between inputs, transformation activities and outputs in order to achieve the system objectives. With the most structured design, there will be less planning and decision-making in the operations of the system. System control consists of all actions necessary to ensure that activities conform to preconceived plans or goals.

Transformation and Value Adding Activities

The objective of combining resources under controlled conditions is to transform them into goods and services having a higher value than the original inputs. The transformation process applied will be in the form of technology to the inputs. The effectiveness of the production factors in the transformation process is known as productivity. The productivity refers to the ratio between values of output per work hour to the cost of inputs. The firms overall ratio must be greater than 1, then we can say value is added

to the product. Operations manager should concentrate improving the transformation efficiency and to increase the ratio.

Fig. 1.1 The conversion process (Schematic model for operations system)



The essence of the operations function is to add value during the transformation process. Value added is the term used to describe the difference between the costs of inputs and the value or price of outputs. Typical examples are given in the table 1 below.

Table 1.1 Input transformation output relationship for typical systems

System	Primary input	Resources	Primary transformation	Desired output
Hospital	Patient	MDS, nurses medical supplies, equipment, bed etc	Health care (physiological)	Healthy individuals
Restaurant	Hungry customer	Food, chief waiters	Well prepared well served food	Satisfied customers
Automobile factory	Sheet steel engine parts	Tools, equipment, workers	Fabrication and assembly of car	High quality cars
College University	High school graduate	Teachers, books, class rooms	Imparting knowledge and skills, information	Educated individuals
Department Store	Shoppers	Displays, stocks of goods, sales, clerks	Attract shoppers promote products	Sales to satisfied customers

WHY STUDY OM?

We study OM for four reasons:

1. OM is one of the three major functions of any organization, and it is integrally related to all the other business functions. All organizations market (sell), finance (account), and produce (operate), and it is important to know how the OM activity functions. Therefore, we study *how people organize themselves for productive enterprise*.

2. We study OM because we want to know *how goods and services are produced*. The production function is the segment of our society that creates the products and services we use.
3. We study OM to *understand what operations managers do*. Regardless of your job in an organization, you can perform better if you understand what operation managers do. In addition, understanding OM will help you explore the numerous and lucrative career opportunities in the field.
4. We study OM *because it is such a costly part of an organization*. A large percentage of the revenue of most firms is spent in the OM function. Indeed, OM provides a major opportunity for an organization to improve its profitability and enhance its service to society.

1.2 HISTORICAL DEVELOPMENT OF OPERATION MANAGEMENT

For over two century's operations and production management has been recognized as an important factor in a country's economic growth. The traditional view of manufacturing management began in eighteenth century when **Adam Smith** recognized the economic benefits of specialization of labour. He recommended breaking of jobs down into subtasks and recognizes workers to specialized tasks in which they would become highly skilled and efficient. In the early twentieth century, F.W. Taylor implemented Smith's theories and developed scientific management. From then till 1930, many techniques were developed prevailing the traditional view. Brief information about the contributions to manufacturing management is shown in the Table 1.2.

Production Management becomes the acceptable term from 1930s to 1950s. As F.W. Taylor's works become more widely known, managers developed techniques that focused on economic efficiency in manufacturing. Workers were studied in great detail to eliminate wasteful efforts and achieve greater efficiency. At the same time, psychologists, socialists and other social scientists began to study people and human behavior in the working environment. In addition, economists, mathematicians, and computer socialists contributed newer, more sophisticated analytical approaches.

With the 1970s emerge two distinct changes in our views. The most obvious of these, reflected in the new name **Operations Management** was a shift in the service and manufacturing sectors of the economy. As service sector became more prominent, the change from 'production' to 'operations' emphasized the broadening of our field to service organizations. The second, more suitable change was the beginning of an emphasis on synthesis, rather than just analysis, in management practices.

Table 1.1 Historical summary of operations management

Date	Contribution	Contributor
1776	Specialization of labour in manufacturing	Adam Smith
1799	Interchangeable parts, cost accounting	Eli Whitney & others
1832	Division of labour by skill; assignment of jobs by Skill; basics of time study	Charles Babbage
1900	Scientific management time study and work study Developed;	Frederick W.Taylor
1900	Motion of study of jobs	Frank B. Gilbreth
1901	Scheduling techniques for employees, machines Jobs in manufacturing	Henry L. Gantt

1915	Economic lot sizes for inventory control	F.W. Harris
1927	Human relations; the Hawthorne studies	Elton Mayo
1931	Statistical inference applied to product quality: quality control charts	W.A. Shewart
1935	Statistical Sampling applied to quality control: inspection sampling plans	H.F.Dodge & H.G.Roming
1940	Operations research applications in world war II	P.M. Blacker & others
1946	Digital Computer	John Mauchlly and J.P.Eckert
1947	Linear Programming	G.B.Dantzig, Williams & others
1950	Mathematical programming, on-linear and stochastic processes	A.Charnes, W.W.Cooper & others
1951	Commercial digital computer: large-scale computations available	Sperry Univac
1960	Organizational behavior: continued study of people at work	L.Cummings, L.Porter
1970	Integrating operations into overall strategy and policy Computer applications to manufacturing & service, scheduling, and control, Material Requirement Planning	W.Skinner J.Orlicky & G. Wright
1980	Quality and productivity applications from Japan: robotics, CAD-CAM	W.E. Deming & J.Juran

1.3 MANUFACTURING OPERATIONS AND SERVICE OPERATIONS

1.3.1 MANUFACTURING OPERATIONS

Manufacturing is the transformation of raw materials into finished goods for sale, or intermediate processes involving the production or finishing of semi-manufactures.

Manufacturing implies production of a tangible output such as an automobile while service generally implies an act such as a doctor's examination.

1.3.2 SERVICE OPERATIONS

Service is defined as either as Services are deeds, processes, and performances or a service is a time-perishable, intangible experience performed for a customer acting in the role of a co-producer. Service enterprises are organizations that facilitate the production and distribution of goods, support other firms in meeting their goals, and *add value* to our personal lives.

Manufacturing and services are often similar in terms of what is done but different in terms of how it is done. For example, both involve design and operating decisions. Manufacturers must decide how large a factory is needed and service organizations (e.g. hospitals) must decide how large a building they need. Both must make locations decision and both involved in scheduling and controlling operations and allocating scarce resources. Most of the differences between manufacturing and service organizations relate to manufacturing being product-oriented and service being act-oriented.

Following characteristics can be considered for distinguishing Manufacturing Operations with Service Operations:

1. The nature and customer contact
2. Uniformity of input
3. Labor content of jobs
4. Uniformity of output
5. Measurement of productivity

1) The nature and customer contact

Service involves a much higher degree of customer contact than manufacturing does. The performance of a service typically occurs at the point of consumption. That is, **the two often occurs simultaneously**. On the other hand, manufacturing allows a separation of production and consumption, so that manufacturing often occurs in an isolated environment away from the customers. Service operations because of their contact with customers can sometimes be much more limited in their range of options in these areas. The product oriented operations can build up inventories of finished goods, which enable them to absorb some of the shocks caused by varying demand. However, service operations cannot build up inventories of time, so service capacity is much more sensitive to demand variability

2) Uniformity of inputs

Services operations are subject to more variability of inputs than manufacturing operations. There is often the ability in manufacturing to carefully control the amount of variability of inputs, so it is often possible to achieve low variability. Consequently, job requirement for manufacturing are generally more uniform than for services.

3) Labor content of Jobs.

Because of on the site consumption of services and because of the high degree of variation of output, service requires high labor content where as manufacturing can be more capital intensive (i.e. mechanized)

4) Uniformity of output

High mechanizations generates products with low variability, so manufacturing tends to be smooth and efficient; service activities sometimes appear to be slow and awkward, and output is more variable.

5) Measurement of productivity

Measurement of productivity is relatively straight forward in manufacturing due to the high degree of uniformity of most manufactured items. However, in many cases variations in demand intensity as well as variations in service requirement from job to job, make productivity measurement considered by more difficult. For example, the work load of the two doctors might be compared. One may have had a large number of routine cases while the other did not, so their productivity would appear to differ unless a very careful analysis is made.

6) Quality Assurance

Quality assurance is more challenging in services when production and consumption occurs at the same time. Moreover, the higher variability of input creates additional opportunity for the quality of output to suffer unless quality assurance is actively managed. Quality at the point of creation is typically more important for services than for manufacturing, where errors can be corrected before the customer receives the outputs.

Generally the differences between manufacturing and service are:

<i>Characteristics</i>	<i>Manufacturing</i>	<i>service</i>
Output	Tangible	Intangible
Customer contact	Low	High
Labor content	Low	High
Uniformity of output	High	Low
Measurement of productivity	Easy	Difficult
Storage	Output can be inventoried	Not

1.4 OPERATIONS DECISION MAKING

Thousand of business decisions are made every day. Not all the decisions will make or break the organization. But each one adds a measure of success or failure to the operations. Even minor decisions determine the company's success or failure. It ranges from simple judgmental to complex analysis which can also involve judgment (past experience & common sense). They involve a way of blending objective and subjective data to arrive at a choice. The use of quantitative methods of analysis adds to the objectivity of such decisions.

Operations Management Decisions

The major areas in which operations managers make decisions are:

- ✚ Strategic (long term) decision
- ✚ Tactical (intermediate term) decision
- ✚ Operational planning and control (short term) decision

A) Strategic (long term) decision: It involves high amount, more effort and it is periodical. The strategic decision includes Product design, Process design, and selection and Location decision. The strategic issue usually broad, addressing questions such as

- ✓ How will we make the product?
- ✓ When do we locate the facility or facilities?
- ✓ How much capacity do we need? When should we add more capacity?

Operations management decisions at the strategic level affect the company's long range effectiveness in terms of how it can address its customers' needs. Thus, for the firm to succeed, these decisions must be in alignment with the corporate strategy. Decision made at the strategic level become the fixed conditions or operating constraints under which the firm must operate in both the intermediate and short term.

B) Tactical (intermediate term) decisions

The tactical decision primarily address how to efficiently schedule material and labor within the constraints of previously made strategic decisions. An issue on which OM concentrates on this level includes:

- ✓ How many workers do we need?
- ✓ When do we need them?
- ✓ Should we work over time?
- ✓ When should we have material delivered?
- ✓ Should we have a finished goods inventory?

These tactical decisions, in turn, become the operating constraints under which operations planning and control decisions are made.

C) Operational planning & control (short term) decision.

Management with the respect to operational planning control is narrow and short term. Issues at this level include:

- ✓ What jobs do we work on today or this week?
- ✓ Whom do we assign to what tasks?
- ✓ What jobs have priority?

MANAGEMENT AS A SCIENCE

Management scientists hold that, education, scientific training and experience can improve a person's ability to make decisions. Scientific decision-making rests upon organized principles of knowledge and depends largely upon the collection of empirical data and analysis of the data in a way that repeatable results will be obtained.

The association of management with the scientific method involves drawing objective conclusions from the facts. Facts come from the analysis of data, which must be gathered, compiled and digested into meaningful form, such as graphs and summary statistics. Computers are helpful in these tasks because they can easily store data and us with the more sophisticated and statistical analysis. But not all variables are quantifiable, so decision-makers must still use some value-based judgments in a decision process.

Thus management as a science is characterized by

- ☞ Organized principle of knowledge.
- ☞ Use of empirical data.
- ☞ Systematic analysis of data.
- ☞ Repeatable results.

CHARACTERISTICS OF DECISIONS

Operations decision range from simple judgments to complex analyses, which also involves judgment. Judgment typically incorporates basic knowledge, experience, and common sense. They enable to blend objectives and sub-objective data to arrive at a choice. The appropriateness of a given type of analysis depends on

- ☞ The significant or long lasting decisions,
- ☞ The time availability and the cost of analysis, and
- ☞ The degree of complexity of the decision.

The significant or long lasting decisions deserve more considerations than routine ones. Plant investment, which is a long-range decision, may deserve more thorough analysis. *The time availability and the cost of analysis* also influence the amount of analysis. *The degree of complexity of the decision increases* when many variables are involved, variables are highly independent and the data describing the variables are uncertain.

Business decision-makers have always had to work with incomplete and uncertain data. Fig. 1.2 below depicts the information environment of decisions. In some situations a decision-maker has complete information about the decision variables; at the other extremes, no information is available. Operations management decisions are made all along this continuum.

Complete certainty in decision-making requires data on all elements in the population. If such data are not available, large samples lend more certainty than do small ones. Beyond this, subjective information is likely to be better than no data at all.

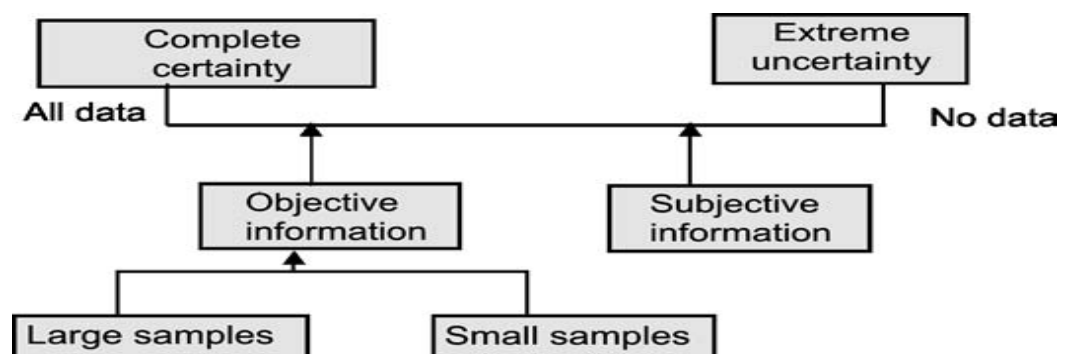


Fig. 1.2 Information continuum

Framework for Decision-Making

An analytical and scientific framework for decision implies the following systematic steps

- ☞ Defining the problem.
- ☞ Establish the decision criteria.
- ☞ Formulation of a model.
- ☞ Generating alternatives.
- ☞ Evaluation of the alternatives.
- ☞ Implementation, monitoring and control.

Defining the Problem

Defining the problem enables to identify the relevant variables and the cause of the problem. Careful definition of the problem is crucial. Finding the root cause of a problem needs some questioning and detective work. If a problem defined is too narrow, relevant variable may be omitted. If it is broader, many tangible aspects may be included which leads to the complex relationships.

Establish the Decision Criteria

Establish the decision criterion is important because the criterion reflects the goals and purpose of the work efforts. For many years profits served as a convenient and accepted goal for many organizations based on economic theory. Nowadays organization will have multiple goals such as employee welfare, high productivity, stability, market share, growth, industrial leadership and other social objectives.

Formulation of a Model

Formulation of a model lies at the heart of the scientific decision-making process. Model describes the essence of a problem or relationship by abstracting relevant variables from the real world situation. Models are used to simplify or approximate reality, so the relationships can be expressed in tangible form and studied in isolation.

Modeling a decision situation usually requires both formulating a model and collecting the relevant data to use in the model. Mathematical and statistical models are most useful models for understanding the complex business of the problem.

Generating Alternatives

Alternatives are generated by varying the values of the parameters. Mathematical and statistical models are particularly suitable for generating alternatives because they can be easily modified. The model builder can experiment with a model by substituting different values for controllable and uncontrollable variable.

Evaluation of the Alternatives

Evaluation of the alternatives is relatively objective in an analytical decision process because the criteria for evaluating the alternatives have been precisely defined. The best alternative is the one that most closely satisfies the criteria. Some models like LPP model automatically seek out a maximizing or minimizing solution. In problems various heuristic and statistical techniques can be used to suggest the best course of action.

Implementation and Monitoring

Implementation and monitoring are essential for completing the managerial action. The best course of action or the solution to a problem determined through a model is implemented in the business world. Other managers have to be convinced of the merit of the solution. Then the follow-up procedures are required to ensure about appropriate action taken. This includes an analysis and evaluation of the solution along with the recommendations for changes or adjustments.

1.5 PRODUCTIVITY MEASUREMENT

Productivity is defined in terms of utilization of resources, like material and labour. In simple terms, productivity is the ratio of output to input. For example, productivity of labour can be measured as units produced per labour hour worked. Productivity is closely linked with quality, technology and profitability. Hence, there is a strong stress on productivity improvement in competitive business environment.

Productivity can be improved by (a) controlling inputs, (b) improving process so that the same input yields higher output, and (c) by improvement of technology. Productivity can be measured at firm level, at industry level, at national level and at international level.

Modern Dynamic Concept of Productivity

Productivity can be treated as a multidimensional phenomenon. The modern dynamic concept of productivity looks at productivity as what may be called “**productivity flywheel**”. The productivity is energized by competition. Competition leads to higher productivity, higher productivity results in better value for customers, and these results in higher share of market for the organization, which results in still keener competition. Productivity thus forms a cycle, relating to design and products to satisfy customer needs, leading to improved quality of life, higher competition *i.e.* need for having still higher goals and higher share of market, and thereby leading to still better designs.

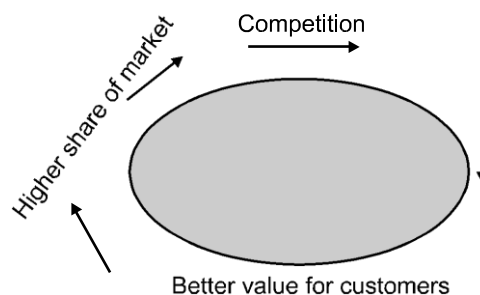


Fig. 1.8 Dynamic concept of productivity

Factor Productivity and Total Productivity

When productivity is measured separately for each input resource to the production process it is called *factor productivity* or *partial productivity*. When productivity is measured for all the factors of production together, it is called *total factor productivity*.

Generally **factor productivity** calculations are required at **firm level and industry level**, whereas total factor productivity calculations are made for measuring productivity at **national and international level**.

Productivity of materials can be measured as output units per unit material consumed. It can also be measured in terms of value generated per unit expenditure in materials.

For measuring productivity of different groups of operatives, different ratios can be used, which are

indicative of output/input relationship. For example, the productivity of assembly line work can be measured as output units per man-hour or alternatively, the value of good produced per cost of labour on assembly line.

One of the primary responsibilities of an operation manger is to achieve productive use of resources. Productivity measures the relationship between output (goods and Services) and inputs (labor, capital, materials, or other resources) used to produce them. Productivity usually expressed the ratio of quantity of output to quantity of input.

$$\text{Thus, Productivity} = \frac{\text{Output}}{\text{Input}}$$

Productivity may be expressed as **partial measures**, **multifactor measures**, or **total measures**.

i. Partial productivity (A single input)

$$\text{Productivity} = \frac{\text{Output}}{\text{Labor}} \text{ or } \frac{\text{Output}}{\text{Capital}} \text{ or } \frac{\text{Output}}{\text{Materials}} \text{ or } \frac{\text{Output}}{\text{Energy}} \text{ \& so on.}$$

ii. Multifactor productivity (Based on more than one input)

It reflects a combination of some or all of the resources used to obtain a certain output.

$$\text{Productivity} = \frac{\text{Output}}{\text{Labor} + \text{Capital}} \text{ or } \frac{\text{Output}}{\text{Labor} + \text{Capital} + \text{Material}}$$

iii. Total productivity (Based on total measure or on all inputs)

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}} = \frac{\text{Goods or services produced}}{\text{All inputs used to produce them}}$$

Example 1.

Three employees process 600 insurance policies in a week. They work 8 hrs. per day, 5-days per week. Find labour productivity.

Solution:

$$\text{Labour Productivity} = [\text{Policies Issued}]/[\text{Employee Hours}]$$

$$P_{\text{labor}} = 600 \text{ policies}/[(3 \text{ employees})(40 \text{ hrs/employee})]$$

$$P_{\text{labor}} = \underline{5 \text{ Policies/hr.}}$$

Example 2.

A team of workers make 400 units of a product, which is valued by its standard cost of 10 birr each (before markups for other expenses and profit). The accounting department reports that for this job the actual costs are:

400 birr for labour,

1000 birr for materials and

300 birr for overhead.

Calculate multi-factor productivity.

Solution:

Multi-Factor Productivity = [Quantity at standard cost]/[Labour cost + Materials cost
+ Overhead cost]

$$P_{mf} = [400 \text{ Units} \times 10] / [400 + 1000 + 300] = 4000 / 1700$$

$$P_{mf} = \underline{2.35}$$

Example 3.

Azim Title company has a staff of 4 each working 8 hours/day (for a payroll cost of 640 birr/day) and overhead expenses of 400 birr/day, Azim process and closes on 8 titles each day. The company recently purchased a computerised title-search system that will allow the processing of 14 titles/day, although the staff, their work hours, and pay will be the same, the overhead expenses are now 800 birr/day.

$$\text{Labour-productivity with the old system} = \frac{8 \text{ titles/day}}{32 \text{ lab.hrs./day}} = 0.25 \text{ titles/lab.hrs.}$$

$$\text{Labour-productivity with the new system} = \frac{14 \text{ titles/day}}{32 \text{ lab.hrs./day}} = 0.4375 \text{ titles/lab.hrs.}$$

$$\text{Multi-factor productivity with the old system} = \frac{8 \text{ titles/day}}{(640 + 400) \text{ birr/day}} = 0.0077 \text{ titles/birr}$$

$$\text{Multi-factor productivity with the new system} = \frac{14 \text{ titles/day}}{(640 + 800) \text{ birr/day}} = 0.0097 \text{ titles/birr}$$

Labour productivity has increased from 0.25 to 0.4375.

The change is 75% increase in labour productivity.

Multi-factor productivity has increased from 0.0077 to 0.0097.

This change is a 25.9% increase in multi-factor productivity.

Example 4.

Productivity can be measured in a variety of ways, such as labour, capital, energy, material usage, and so on. At Modern Lumber, Inc. Ali Caliskan, president and producer of apple crates sold to growers, has been able, with his current equipment, to produce 240 crates per 100 logs, the current purchases 100 logs per day and each log requires 3 labour-hrs to process.

He believes that he can hire a professional buyer who can buy a better-quality log at the same cost. If this is the case, he can increase his production to 260 crates/100 logs, this labour-hours will increase by 8 hrs per day.

What will be the impact on productivity (measured in crates per labour-hour) if the buyer is hired?

- a. Ali Caliskan has decided to look at his productivity from a multifactor (total factor productivity) perspective. To do so, he has determined his labour, capital, energy and material usage and decided to use money units (MU for dollars) as the common denominator

His total labour-hours are now 300 hrs/day and will increase to 308 hrs/day. His capital and energy costs will remain constant at 350 birr and 150 birr per day, respectively. Material costs for the 100 logs per day are 1000 birr and will remain the same.

Because he pays an average of 10 birr/hr (with fringes), Caliskan wants to determine his productivity increase?

Solution:

$$\text{Current Labour Productivity} = \frac{240 \text{ crates}}{100 \text{ logs} \times 3 \text{ hrs}} = 0.8 \text{ crates/lab-hr}$$

$$\text{Labour Productivity with buyer} = \frac{260 \text{ crates}}{(100 \text{ logs} \times 3 \text{ hrs}) + 8 \text{ hrs}} = 0.844 \text{ crates/lab-hr.}$$

Using current productivity (0.8 from (a)) as a base, the increase will be 5.5% or a 5.5% increase)

a. Current System

Labour 300hrs@10 birr	=3000
Material 100 logs/day	= 1000
Capital	350
Energy	<u>150</u>
Total Cost	<u>4.500 birr</u>

System with Professional Buyer

308hrs@10 birr	= 3080 birr
	1000
	350
	<u>150</u>
	<u>4.580 birr</u>

$$\text{Productivity of current system} = \frac{240 \text{ crates}}{4500} = 0.0533$$

$$\text{Productivity of proposed system} = \frac{260 \text{ crates}}{4580} = 0.0567$$

Using current productivity (0.0533) as a base, the increase will be 0.047. That is, 6.4% increase.

Example 5:

Ilhan Bal makes wooden boxes in which to ship bikes. Ilhan and his three employees invest 40 hours per day making the 120 boxes.

- What is their productivity?
- Ilhan and his employees have discussed redesigning the process to improve efficiency. If they can increase the rate to 125 per day. What would be their new productivity?
- What would be their *increase* in productivity?

Solution:

- $P_{\text{labour}} = \text{output/input} = 120 \text{ boxes}/40 \text{ hours} = \underline{3.0 \text{ boxes/hour}}$
- $P_{\text{labour}} = \text{output/input} = 125 \text{ boxes}/40 \text{ hours} = \underline{3.125 \text{ boxes/hour}}$
- Change in productivity = 0.125 boxes/hour
Percentage change = $0.125 \text{ boxes/hour} / 3.0 \text{ boxes/hour} = \underline{4.166\%}$

Example 6.

Magusa Metal Works produces cast bronze valves on a 12 person assembly line. On a recent day, 240 valves produced during an 8 hour shift. Calculate the labour productivity.

Solution:

Total labour hours = 12 persons @ 8 hours = 96 hours

Labour productivity = 240 valves/96 hours = 2.5 valves/hour

Example 7.

Gaye produces “Final Exam Care Packages” for resale by the sorority. She is currently working a total of 6 hours a day to produce 120 care packages.

- a. What is Gaye’s productivity?
- b. Gaye thinks that by redesigning the package she can increase her total productivity to 150 care packages per day. What would be her new productivity?
- c. What will be the increase in productivity if Gaye makes the change?

Solution:

- a. $P = \text{units produced}/\text{input} = 120 \text{ pkgs}/6\text{hrs} = \underline{20 \text{ pkgs/hr}}$
- b. $P = \text{units produced}/\text{input} = 150 \text{ pkgs}/6\text{hrs} = \underline{25 \text{ pkgs/hr}}$
- c. Increase in productivity = $\{25 \text{ pkgs/hr} - 20 \text{ pkgs/hr}\}/20 \text{ pkgs/hr} = \underline{25 \%}$

Productivity comparisons can be made in two ways:

- First, a company can compare itself with similar operations within its industry or it can use industry data when such data are available. For example, comparing productivity among the different stores in a franchise.
- Second, to measure productivity overtime with in the same operations. I.e. A Company can compare its productivity in one time period with that of the next.

Productivity Analysis

For the purposes of studies of productivity for improvement purposes, following types of analysis can be carried out:

1. **Trend analysis:** Studying productivity changes for the firm over a period of time.
2. **Horizontal analysis:** Studying productivity in comparison with other firms of same size and engaged in similar business.
3. **Vertical analysis:** Studying productivity in comparison with other firms of different sizes in the same industry.
4. **Budgetary analysis:** Setting up a norm for productivity for a future period as budget, based on studies as above, and planning strategies to achieve it.

Factors Affecting Productivity

Numerous factors affect productivity. However some of the principal factors influencing productivity are:

1. **Capital/labour ratio:** It is a measure of whether enough investment is being made in plant, machinery, and tools to make effective use of labour hours.
2. **Scarcity of some resources:** Resources such as energy, water and number of metals will create productivity problems.
3. **Work-force changes:** Change in work-force effect productivity to a larger extent, because of the labour turnover.
4. **Innovations and technology:** This is the major cause of increasing productivity.
5. **Managerial factors:** Managerial factors are the ways an organization benefits from the unique planning and managerial skills of its manager.
6. **Quality of work life:** It is a term that describes the organizational culture, and the extent to which it motivates and satisfies employees.

KEY POINTS FOR IMPROVING PRODUCTIVITY

There are a number of key steps that a company or a department can take toward improving productivity:

- 1) Develop productivity measures for all operations. Measurement is the first step in managing and controlling the operation.
- 2) Look at the system as a whole in deciding which operations to concentrate on; it is overall productivity that is important.
- 3) Develop methods for achieving productivity improvements, such as soliciting ideas from workers, studying how other firms have increased productivity and reexamining the way work is done.
- 4) Establish reasonable goals for improvement.
- 5) Make it clear that management supports and encourage productivity, consider incentives towards for contributions.
- 6) Measure improvements and publicize them.

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CHAPTER TWO

OPERATIONS STRATEGY & COMPETITIVENESS

2.1 INTRODUCTION TO OPERATIONS STRATEGY

Strategies: are plans for achieving goals. The organization strategy provides the overall direction for the organization. It is broad in scope, covering the entire organization. Operations strategy is narrower in scope, dealing primarily with the operations aspect of the organization.

Relationship between Operations and Organizational Strategy

Organizational strategy is

- ☞ An overall big picture for the whole organization.
- ☞ Longer in time horizon
- ☞ Less detailed and broader in scope.

Operations Strategy is

- ☞ Narrower in scope and in more detail
- ☞ Prepared by middle management.
- ☞ Should be in line with the Organization strategy

Operations Strategy relates to products, processes, methods, operating resources quality, costs, lead times, and scheduling.

An operations strategy is a strategy for the operations functions that is linked to the business strategy and other functional strategy, leading to a consistence pattern of decision making and competitive advantage for the firm.

An operations strategy consists of a sequence of decisions that, over time, enables a business unit to achieve a desired operations structure, infrastructure, and set of specific capabilities in support of the competitive priorities.

Operations strategy: *Operations strategy is concerned with setting broad policies and plans for using the resources of the firm to best support the firm's long-term competitive strategy.* It has a long-term impact on the **nature and characteristics** of the organization. In large measure, strategies affect the ability of an organization to compete.

Operations strategy concerns the pattern of strategic decisions and actions which set the role, objectives and activities of the operation. The term ‘operations strategy’ sounds at first like a contradiction. How can ‘operations’, a subject that is generally concerned with the day-to-day creation and delivery of goods and services, be strategic? ‘Strategy’ is usually regarded as the opposite of those day-to-day routine activities. But **‘operations’ is not the same as ‘operational’**. ***‘Operations’ are the resources that create products and services. ‘Operational’ is the opposite of strategic, meaning day-to-day and detailed.*** So, one can examine both the operational and the strategic aspects of operations. It is also conventional to distinguish between the **‘content’ and the ‘processes** of operations strategy.

The content of operations strategy is the specific decisions and actions which set the operations role, objectives and activities. The process of operations strategy is the method that is used to make the specific ‘content’ decisions. Nor is there universal agreement on how an operations strategy should be described.

OPERATIONS COMPETITIVE DIMENSIONS

Given the choices customers face today, how do they decide which product or service to buy? ***Different customers are attracted by different attributes.*** Some customers are interested primarily in the **cost** of a product or service and, correspondingly, some companies attempt to **position** themselves to offer the lowest prices. Companies must be competitive to sell their goods and services in the market place. Business organizations compete with one another in variety of ways i.e. by identifying **operating priorities**. The major competitive dimensions (operations priorities) that form the competitive position of a firm include the following:

1. **Cost:** ***“Make the product or deliver the service cheap.”*** within every industry, there is usually a segment of the market that buys strictly on the basis of **low cost**. To successfully compete in this niche, a firm must be the low-cost producer, but even doing this not always guarantee profitability and success.
2. **Quality:** ***“Make a great product or deliver a great service.”*** There are two characteristics of a product or service that define **quality: design quality and process quality**. ***Design quality relates to the set of feature the product or service contains.*** This relates directly to the design of the product or service. The goal in establishing the proper level of design quality is to focus on the requirements of the customer. Overdesigned products and service with too many or inappropriate features will be viewed

as prohibitively expensive. In comparison, under designed products and services will lose customers to products that cost a little more but are perceived by customers as offering greater value. *Process quality* is critical because it relates directly to reliability of the product or service. Thus, **the goal of process quality is to produce defect-free product and service.**

3. Delivery speed/timeliness: “*Make the product or deliver the service quickly.*” The elapsed time between customers requesting products or services and receiving them. A company that can offer an on- site repair service in only 1 or 2 hours has a significant advantage over a competing firm that guarantees service only within 24 hours. **The three delivery speed priorities are:**

- a) **Fast delivery time:** delivery time is the **elapsed** time between receiving a customer order and filling it. Sometime it is known as **lead time**.
- b) **On time delivery:** measures the frequency in which delivery promise is met.

Delivery Reliability: “*Deliver it when promised.*” This dimension relate to the firm’s ability to supply the product or service on or before a promised delivery due date.

- c) **Development speed:** measures how quickly a new product is introduced covering from idea generation to commercialization. It is important especially in the fashion applied industry.

4. Flexibility: the ability to respond to changes. The better a company as able to respond to changes, the greater the competitive advantage over another company that is not able to respond. Flexibility could be pursuing through value adding system. The two types of flexibility priorities are:

- i. **Customization:** the ability to satisfy the unique needs of each customer by changing a product design.
- ii. **Volume flexibility:** the ability to accelerate or decelerate the rate of production quickly to handle fluctuations and demand.

Order Winners and Qualifiers: The Marketing – Operations Link

An interface between marketing and operations is necessary to provide a business with an understanding of its market from both perspectives. Terry Hill, a professor at Oxford University, has coined the terms *order winner* and *order qualifier* to describe marketing oriented dimensions that key to competitive success.

Order Winner: An order winner is a criterion that differentiates the products or services of one firm from another. Depending on the situation, the order winning criterion may be the cost of the product (price), product quality and reliability, or any of the other dimensions. Order winners can be **defined as the**

characteristics of a firm that distinguish it from its competition so that it is selected as the source of purchase.

Order qualifier: An order qualifier is a screening criterion that permits a firm's products even be considered as possible candidates for purchase. **Order-qualifiers can be defined as the minimum elements or characteristics that a firm or its products must have to even be considered as a potential supplier or source.**

In Europe, for example, the vast majority of companies today require that their vendors be **ISO-9000** certified. (This certification ensures that a firm has documented all of its processes.) Thus, **ISO-9000 certification is an order-qualifier in Europe**. In contrast, most companies in Canada at this time are not ISO-9000 certified. As a consequence, ISO-9000-certified companies in Canada use their certification as an order-winner (that is, ISO-9000 certification distinguishes them as being better than their competition).

It is important to remember that the order-winning and order-qualifying may change overtime. Basically, when very few firms offer a specific characteristic, such as high quality, customization, or outstanding service, that characteristic can be defined as an order-winner. However, over time, as more and more firms begin to offer that same enhancement, the order-winner becomes an order-qualifier. In other words, it becomes the minimum acceptable level for all competitors. As a result, the customer uses some other new enhancement or characteristic to make the final purchase.

DEVELOPING OPERATIONS STRATEGY

An operations strategy is a strategy for the operations functions that is linked to the business strategy and other functional strategy, leading to a consistence pattern of decision making and competitive advantage for the firm. An operations strategy consists of a sequence of decisions that, over time, enables a business unit to achieve a desired operations structure, infrastructure, and set of specific capabilities in support of the competitive priorities.

Elements of operations strategy

Mission: describes the purposes of the organization, the **reason for its existence**.

Objectives: As noted previously the objectives of operations are cost, quality, delivery and flexibility. These objectives should be **derived from the mission**, and they constitute a **restatement of the mission in quantitative and measurable terms**.

Policies: Policies should indicate how the operations objective will be achieved.

Distinctive competence: distinctive competence **is something that operations do better than anyone else, and are those especial attributes or abilities possessed by organizations that give it a competitive edge.** That means something that is differentiated from its competitors.

Strategy can be considered to exist at **three levels** in an organization

a) Corporate strategy: defines what business the company pursues. Corporate level strategy is the **highest level of strategy**. It sets the **long-term direction and scope** for the whole organization. If the organization comprises more than one business unit, corporate level strategy will be concerned with what those businesses should be, how resources (e.g. cash) will be allocated between them, and how relationships between the various business units and between the corporate centre and the business units should be managed. **Organizations often express their strategy in the form of a corporate mission or vision statement.**

Key issues

- ☞ What businesses shall we be in?
- ☞ What businesses shall we acquire or divest?
- ☞ How do we allocate resources between businesses?
- ☞ What is the relationship between businesses?
- ☞ What is the relationship between the centre and the businesses?

b) Business strategy: follows from the corporate strategy and defines how a particular business will compete or how to become better within industry. Business level strategy is primarily concerned with how a particular business unit should compete within its industry, and what its strategic aims and objectives should be. Depending upon the organization's corporate strategy and the relationship between the corporate centre and its business units, a business unit's strategy may be constrained by a lack of resources or strategic limitations placed upon it by the centre. **In single business organizations, business level strategy is synonymous with corporate level strategy.** It can be in the form of:

- **Customer intimacy:** This strategy requires excellent customer management process such as relationship management and solution development. A customer intimate company builds bonds with its customers. It knows the people to whom it sells and the product and services it needs.
- **Product leadership:** Product leadership strategy would require a leading edge innovation process that creates new products with best in class functionality and brings them to market rapidly.
- **Operational excellence strategy:** this strategy emphasizes cost, quality, quickness of operating process, excellent supplier relationships, and speed and efficiency supply and distribution process.

c) **Functional level strategy:** The bottom level of strategy is that of the individual function (operations, marketing, finance, etc.). These strategies are concerned with how each function contributes to the business strategy, what their strategic objectives should be and how they should manage their resources in pursuit of those objectives.

The two diametrically opposite business strategies are:

A. *Product imitator (operational excellence) business strategy*

This would be a typical of a mature, price sensitive market with standardized product. In this case, the operations mission would emphasize cost as the dominant objective and operations should strive to reduce costs through such policies as superior technology, low personnel cost, low inventory levels, a high degree of integration, and quality assurance aimed at saving costs.

B. *Product innovation and new product introduction (product leadership)*

This strategy would typically be used in an emerging and possibly growing market where advantages can be gained by bringing out superior-quality product in short amount of time. In this case, operations would emphasize flexibility to rapidly and effectively introduce superior new product as its mission.

Criteria for Evaluating an Operations Strategy

Consistency (is the strategy consistent...?)

- ☞ Between the operations strategy and the overall business strategy
- ☞ Between the operations strategy and the other functional strategies within the business
- ☞ Between the different decision areas of operations strategy
- ☞ Between the operations strategy and the business environment (resources available, competitive behavior, governmental restraints, etc)

Contribution to competitive advantage (Does the strategy ...?)

- ☞ Enable operations to set priorities that enhance the competitive advantage
- ☞ Directing attention to opportunities that complement the business strategy
- ☞ Promoting clarity regarding the operations strategy throughout the business unit so its potential can be fully realized
- ☞ Providing the operations capabilities that will be required by the business in the future

Model of the strategic role of operations

Hayes and Wheelwright's four stage model is underpinned by their belief that an **organization's operations can provide a source of competitive advantage**. It can only do this if the operations function is managed strategically. As such, they argue, all organizations should aspire to reach the highest level possible, ultimately reaching stage 4. The four-stage models of the strategic role of operations are discussed below.

Stage 1 Internally Neutral: The operations function is **internally focused and reactive**. They are viewed as a '**necessary evil**'. The best that the organization hopes for is that operations '**don't screw up**'. A stage 1 organization finds it **impossible to manage its operations strategically**, as its operations performance objectives are **continually changing** between low cost, increased flexibility, improved quality, etc. **Because** operations managers never have the time to focus on a consistent set of objectives, a stage 1 organization is characterized by a **reactive approach** to operations management. In such an organization, **operations can never provide a source of competitive advantage**.

Stage 2 Externally Neutral: The operations function **tries to be as good as the competition**, or to achieve **parity with industry norms**. Such an organization is likely to **benchmark its operations against its competitors**, and **adopt best practice** in its industry so that it does not hold the organization back.

A stage 2 organization manages its operations by seeking to **emulate those of its competitors**. It is likely to **copy the prevailing best** practices of its industry, such as JIT (just-in-time), TQM (total quality management), BPO (business process outsourcing) etc. However, as they always adopt these techniques in the **wake of industry leaders**, they are never likely to have developed the same level of expertise in their application. The best that such an approach can achieve is to match the operations performance of its competitors. Although the combination of operations practices adopted by a stage 2 organization may be considered by some as amounting to an operations strategy in that they are consistent, they will not be overtly linked to business strategy.

Indeed, it may be that such an operations strategy is inappropriate for the organization's business strategy. In any event, **a stage 2 organization's operations cannot provide the basis for competitive advantage**.

Stage 3 Internally Supportive: The operations function seeks to provide credible support for the organization's business strategy. An operations strategy will be developed which will be derived from, and support, the business strategy. **The organization's operations are likely to be amongst the best in its industry**.

A stage 3 organization has an operations strategy that is linked to and derived from its business strategy. This means that its **operations performance objectives are aligned with**, and supportive of, its business objectives, **offering the possibility that operations can provide the means of achieving a competitive advantage**. The chances of achieving competitive advantage will be considerably increased if the organization has adopted industry best practice in its operations.

Stage 4 Externally Supportive: The operations function **provides the basis of competitive advantage** for the organization, by **setting the standard in their industry**. The operations function is likely to **aim to be world class by seeking to emulate best practice wherever it is to be found**. Operations will be seen as the means of **exceeding customer expectations** by **delighting the customer**. Operations will be managed **proactively to drive** the business strategy of the organization.

A stage 4 organization is radically different to one at any of the other stages. A stage 4 **organization uses its operations excellence as the basis for its business strategy** – an operations-based strategy. The operations of a stage 4 organization are at the forefront of developments in best practice in that they set industry standards in ways that delight customers. Thus, the organization's operations **enable it to retain its existing customers and attract new ones**. For an operations-based **competitive advantage to be sustainable**, the organization must continually develop its operations, as any source of advantage is liable to be imitated by competitors. **To remain at stage 4, an organization needs to learn how to make the most of its existing resources and competences to learn how to develop new capabilities**. Recent advances in the understanding of organizational performance have emphasized the importance of path dependency (i.e. how organizations got to their present position), the dynamic nature of the capabilities on which organizational success ultimately depends and the role of organizational learning.

2.2 OPERATIONS STRATEGY IN MANUFACTURING

Operation strategy cannot be designed in vacuum. It must be linked ***vertically to the customer*** and ***horizontally to other parts of the enterprise***.

Developing a manufacturing strategy

The main objectives of manufacturing strategy development are:

- ☞ To translate required competitive dimensions (typically obtained from marketing) into specific performance requirement for operations and

☞ To make the plan necessary to ensure that operations (and enterprise) capabilities are sufficient to accomplish them. The steps for prioritizing these dimensions are as follows:

1. Segment the market according to the product group
2. Identify the product requirements, demand patterns, and profit margin of each group.
3. Determine the order winners and order qualifiers for each group.
4. Convert order winners into specific performance requirements.

The process of achieving a satisfactory manufacturing segmentation that maintains focus is often a matter of deciding which products or product groups fit together in the sense that they have similar market performance characteristics or place similar demands on the manufacturing system.

2.3 OPERATIONS STRATEGY IN SERVICES: Operations strategy in service firms is generally inseparable from the corporate strategy. For most services, the service delivery system is the business, and hence any strategic decision must include operations considerations.

CHAPTER THREE

DESIGN OF THE OPERATION SYSTEM

3.1 PRODUCT AND SERVICE DESIGN

Why is good design so important?

Good design satisfies customers, communicates the purpose of the product or service to its market and brings financial rewards to the business. The objective of good design, whether of products or services, is to satisfy customers by meeting their actual or anticipated needs and expectations. This, in turn, enhances the competitiveness of the organization. Product and service design, therefore, can be seen as starting and ending with the customer. So the design activity has one overriding objective: to provide products, services and processes which will satisfy the operation's customers. *Product* designers try to achieve aesthetically pleasing designs which meet or exceed customers' expectations. They also try to design a product which performs well and is reliable during its lifetime. Further, they should design the product so that it can be manufactured easily and quickly. Similarly, *service* designers try to put together a service which meets, or even exceeds, customer expectations. Yet at the same time the service must be within the capabilities of the operation and be delivered at a reasonable cost.

Using design throughout the business ultimately boosts the bottom line by helping create better products and services that compete on value rather than price. *Design* helps businesses connect strongly with their customers by anticipating their real needs. That in turn gives them the ability to set themselves apart in increasingly tough markets. Furthermore, using design both to generate new ideas and turn them into reality allows businesses to set the pace in their markets and even create new ones rather than simply responding to the competition.

The essence of any organization is the products or services it offers. There is an obvious link between the design of those products or services and the success of the organization that have well-designed products or services. Firms that have well-designed products or services are more likely to realize their goals than those with poorly designed products or services. Hence, organizations have a vital stake in a good product and service design. This unit presents the major aspects of product and services design and how the process to produce them is designed and selected.

Product and service design plays a strategic role in the degree to which an organization is able to achieve its goals. It is a major factor in customer satisfaction, product and service quality, and production costs (price). Because product design is concerned with the functional (use) and aesthetic requirements

necessary to meet the demands of the market place and at the sometime achieve an acceptable rate of return. Decisions related to product designs have far reaching effect on the future of an organization. Therefore, great care must be taken while designing a product/service.

The objectives of product and service design may vary from situation to situation. Generally, however, the objectives/reasons are:

1. To introduce new or revised products or service to the market as quickly as possible;
2. To design product or service that have customer appeal;
3. To increase the level of customer satisfaction;
4. To reduce costs and;
5. To increase quality

In a competitive environment getting new or improved products or services to the market a head of competitors gives an organization a competitive advantage that can lead to increased profits as well increased market share and can create an image of the organization as a leader.

***Product design** defines a product's characteristics, such as its appearance, the materials it is made of, its dimensions and tolerances, and its performance standards.*

***Service design** defines the characteristics of a service, such as its physical elements, and the esthetic and psychological benefits it provides.*

Service design is different from product design in that we are designing both the service and the entire *service concept*. As with a tangible product, the service concept is based on meeting customer needs. The service design, however, adds to this the esthetic and psychological benefits of the product. These are the service elements of the operation, such as promptness and friendliness. They also include the ambiance, image, and “feel-good” elements of the service.

PHASES OF PRODUCT DESIGN

The three major functions involved in product and service design are marketing, product development and manufacturing. **Marketing** has the responsibility for suggesting ideas for new product and for providing product specification for existing product lines. **Product development** has the responsibility for moving the technical concept for the product to its final design and **manufacturing/operation** function has the responsibility for selecting and/or configuring the process by which the product is to be manufactured.

The product development activity provides the link between the customer needs and expectations and the activity required to manufacturing the product. This will take you to the discussion of new product design processes.

The design process begins with motivation for design. For a new business or a new product, the motivation may be obvious to achieve the goals of the organization and realize new opportunities. However, making them happen /utilized them is a demanding challenge. New product development entails a complex set of activities that cut across most functions in business.

There are also specific external factors to consider, such as government regulations, competitive pressures, customer needs, the appearance of new technologies etc. Ultimately, the customer is the challenging force for product and service design.

STEPS IN PRODUCT DESIGN

Certain steps are common in the development of most product designs. They are the following:

1. IDEA DEVELOPMENT

For product designs and development the starting point is an idea. Ideas for new products and services should be sought from a variety of sources including market research, customer view points, the organization's research and development (R&D) department and competitors. The idea might come from a product manager who spends time with customers and has a sense of what customers want. To remain competitive, companies must be innovative and bring out new products regularly.

A. Ideas from Customers

The first sources of ideas are customers, the driving force in the design of products and services. Marketing is a vital link between customers and product design. Market researchers collect customer information nby studying customer buying patterns and using tools such as customer survey sand focus groups. Analyzing customer preferences is an ongoing process. Customer preferences next year may be quite different from what they are today. For this reason, the related process of forecasting future consumer preferences is important, though difficult.

B. Competitors as a source

Competitors are another source of ideas. A company learns by observing itscompetitors' products and services and the success rate of these products and services.This includes looking at product design,

pricing strategy, and other aspects of the operation. Studying the practices of companies considered “best in class” and comparing the performance of our company against theirs is called **benchmarking**. We can benchmark against a company in a completely different line of business and still learn from some aspect of that company’s operation. **Benchmarking** is the process of studying the practices of companies considered “best in class” and comparing your company’s performance against theirs.

Reverse Engineering is another way of using competitors’ ideas is to buy a competitor’s new product and study its design features. Using a process called **reverse engineering**; a company’s engineers carefully disassemble the product and analyze its parts and features. This approach was used by the Ford Motor Company to design its Taurus model. Ford engineers disassembled and studied many other car models, such as BMW and Toyota, and adapted and combined their best features. Other sources of ideas are company’s R & D department, suppliers, the company’s employees, and new technological developments.

2. PRODUCT SCREENING

After a product idea has been developed it needs to be evaluated to determine its likelihood of success. This is called *product screening*. The company’s product screening team evaluates the product design idea according to the needs of the major business functions. The screening process consists of market analysis, economic analysis and technical analysis.

A. Market analysis

Market analysis consists of evaluating the product concept with potential customers through interviews, focus groups and other data collection methods. The physical product may be tested by supplying a sample for customer evaluation. The market analysis should identify whether sufficient demand for the proposed product exists and its fit with the existing marketing strategy.

B. Economic Analysis

Economic analysis consists of developing estimates of production and demand costs and comparing them with estimates of demand. In order to perform the analysis requires an accurate estimate demand as possible derived from statistical forecasts of industry sales and estimates of market share in the sector of the product is competing in. These estimates will be based on a predicted price range for the product which is compatible with the position of the new product in the market. In order to assess the feasibility of the projected estimates of product cost in terms of such factors as materials, equipment and personnel

must be estimated. Techniques such cost/benefit analysis, decision theory and accounting measures such as net present value (NPV) and internal rate of return may be used to calculate the profitability of a product. Another tool that can be used is the cost-volume-profit model that provides a simplified representation that can be used to estimate the profit level generated by a product at a certain product volume.

C. Technical analysis

Technical analysis consists of determining whether technical capability to manufacture the product. This covers such issues as ensuring materials are available to make the product to the specification required, and ensuring the appropriate machinery and skills are available to work with these materials. The technical analysis must take into account the target market and so the product designers have to consider the costs of manufacturing and distributing the product in order to ensure it can be sold at a competitive price.

3. PRELIMINARY DESIGN AND TESTING

Once a product idea has passed the screening stage, it is time to begin preliminary design and testing. At this stage, design engineers translate general performance specifications into technical specifications. Prototypes are built and tested. Changes are made based on test results and the process of revising, rebuilding a prototype, and testing continues. For service companies this may entail testing the offering on a small scale and working with customers to refine the service offering.

4. FINAL DESIGN

Following extensive design testing the product moves to the final design stage. Few ideas will reach the final product selection stage. This is where final product specifications are drawn up. The final specifications are then translated into specific processing instructions to manufacture the product, which include selecting equipment, outlining jobs that need to be performed, identifying specific materials needed and suppliers that will be used, and all the other aspects of organizing the process of product production.

METHODS TO IMPROVE QUALITY OF DESIGN

A. Quality Function Deployment (QFD)

One approach to getting the voice of the customer in to the design specification of product is QFD. This approach which uses inter functional teams from marketing, design engineering, and manufacturing. The

QFD process begins with studying and listening to customers determines the characteristics of a superior product. Through market research the customers' product needs and preferences are defined and broken down into categories called customer requirements. A customer survey and interview can be conducted to determine customer requirement about the product, customers may also be asked to compare and rate the product of the company with product of competitors in the industry. This process will help the company to determine the important characteristics of the product to the customer and to evaluate the company's product with others (competitors) the end and ultimate result is a better understanding and focus on product characteristics that require improvement.

B. Value Analysis/ Value Engineering (VA/VE)

The purpose of value Analysis/Value engineering (VA/VE) is to simplify products and processes. Its objective is to achieve equivalent or better performance at lower cost while maintaining all functional requirements defined by customer; VA/VE does this by identifying and eliminating unnecessary cost, reduced complexity of products, improvement of functional aspect of the product, improved job design and safety, improved maintainability or serviceability of the product. The term VA is more or less the same with the VE. The difference lies only on the timing of them. Technically, VA deals with product already in production process and is used to analyze product specifications and requirements as shown in production developments and purchase requests. Typically, value engineering is considered as cost-avoidance method. VE focus on pre-production design improvement. VE activities are concerned with improvement of design and specifications at research, development, and design stages of product development.

FACTORS TO CONSIDER IN PRODUCT DESIGN

Here are some factors that need to be considered during the product design stage.

I. Design for Manufacture

When we think of product design we generally first think of how to please the customer. However, we also need to consider how easy or difficult it is to manufacture the product. Otherwise, we might have a great idea that is difficult or too costly to manufacture.

Design for manufacture (DFM) is a series of guidelines that we should follow to produce a product easily and profitably. DFM guidelines focus on two issues:

☞ **Design simplification** means reducing the number of parts and features of the product whenever possible. A simpler product is easier to make, costs less, and gives us higher reliability.

☞ **Design standardization** refers to the use of common and interchangeable parts. By using interchangeable parts we can make a greater variety of products with less inventory and significantly lower cost and provide greater flexibility.

II. Concurrent Engineering

Concurrent engineering is an approach that brings many people together in the early phase of product design in order to simultaneously design the product and the process. This type of approach has been found to achieve a smooth transition from the design stage to actual production in a shorter amount of development time with improved quality results.

The old approach to product and process design was to first have the designers of the idea come up with the exact product characteristics. Once their design was complete they would pass it on to operations who would then design the production process needed to produce the product. This was called the “*over-the-wall*” approach, because the designers would throw their design “over-the-wall” to operations who then had to decide how to produce the product.

There are many problems with the old approach. *First*, it is very inefficient and costly. For example, there may be certain aspects of the product that are not critical for product success but are costly or difficult to manufacture, such as a dye color that is difficult to achieve. Since manufacturing does not understand which features are not critical, it may develop an unnecessarily costly production process with costs passed down to the customers. Because the designers do not know the cost of the added feature, they may not have the opportunity to change their design or may do so much later in the process, incurring additional costs. Concurrent engineering allows everyone to work together so these problems do not occur.

A second problem is that the “*over-the-wall*” approach takes a longer amount of time than when product and process design work together. When product and process design work together much of the work is done in parallel rather than in sequence. In today’s markets, new product introductions are expected to occur faster than ever.

The third problem is that the old approach does not create a team atmosphere, which is important in today’s work environment. Rather, it creates an atmosphere where each function views its role separately in a type of “us versus them” mentality. With the old approach, when the designers were finished with the designs, they considered their job done. If there were problems, each group blamed the other. With

concurrent engineering the team is responsible for designing and getting the product to market. Team members continue working together to resolve problems with the product and improve the process.

STRATEGIES FOR NEW PRODUCT INTRODUCTION

There are three fundamentally different ways to introduce new products. These approaches are called market pull, technology push and inter functional.

I. Market pull View

This is an approach when a firm begins product development with a market opportunity and when uses whatever available technologies are required to satisfy the market need. The customer needs are determined, and then the firm organizes the resources need to supply the customer. The market will "pull" through the products that are made.

II. Technology push view

In developing technology push products, a firm begins with a new proprietary technology and looks for an appropriate market in which to apply this technology. The firm should pursue a technology based advantage by developing superior technologies and products. The products are then pushed into the market and marketing job is to create demand for these superior products. Since the products have superior technology, they will have a natural advantage in the market and the customer will want to buy them.

III) Inter functional view

This view holds that the product should not only fit the market needs but have a technological advantage as well. To accomplish this, all functions such as marketing, engineering and finance should cooperate to design the new products needed by the firm. This approach is usually the best, since it includes both market and technological considerations in the new product design, but it is the most difficult to implement.

3.2 PROCESS SELECTION

So far we have discussed issues involved in product design. Though product design is very important for a company, it cannot be done separately from the selection of the process. Among the most important decisions made by operations managers are those involving the design and improvement of the process for producing goods and services. These decisions include choice of process and technology, analysis of flows through operations, and the associated value added in operations. **Process selection decisions**

determine the type of process used to make the product or service. The considerations required for process selection include the *volume of the product* and whether the product is *standardized or customized*. Generally speaking, high volume products that are standardized will be made on an assembly line; while low volume customized products will be made in a batch operation.

Process decisions are *strategic* in nature. They require a *long term* perspective and a great deal of cross functional coordination, since marketing, finance, human resource, and operations issues are all important. Process selection decisions tend to be *capital intensive* and cannot be easily changed. Therefore, the firm is committed to the process choice and bound by these decisions for years to come.

Types of Process classification

Two main types of process classifications are based on *product flow characteristics* and classification by type of *customer order*.

PRODUCT-FLOW CHARACTERISTICS

In manufacturing, the product flow is the same as the flow of materials, since materials are being converted into the product. In services, there might not be product flow, but there would be a flow of customers or information.

All processes can be grouped into two broad categories: **Intermittent** operations and **continuous** operations. These two categories differ in almost every way. Once we understand these differences we can easily identify organizations based on the category of process they use. Dividing processes into two fundamental categories of operations is helpful in our understanding of their general characteristics. To be more detailed, we can further divide each category according to **product volume** and degree of **product standardization** as follows. **Intermittent** operations can be divided into **project** processes and **batch** processes. **Continuous** operations can be divided into **line** processes and **continuous** processes.

1. Intermittent Operations (Process-focused)

Intermittent operations are used to produce many different products with varying processing requirements in lower volumes. Because different products have different processing needs, there is no standard route that all products take through the facility. Instead, resources are grouped by function and the product is routed to each resource as needed. Think about a health-care facility. Each patient, “the product,” is routed to different departments as needed. One patient may need to get an x-ray, go to the lab for blood

work, and then go to the examining room. Another patient may need to go to the examining room and then to physical therapy.

To be able to produce products with different processing requirements, intermittent operations tend to be labor intensive rather than capital intensive. Workers need to be able to perform different tasks depending on the processing needs of the products produced. Often we see skilled and semi-skilled workers in this environment with a fair amount of worker discretion in performing their jobs. Workers need to be flexible and able to perform different tasks as needed for the different products that are being produced. Equipment in this type of environment is more general purpose to satisfy different processing requirements. Automation tends to be less common, because automation is typically product specific. Given that many products are being produced with different processing requirements, it is usually not cost efficient to invest in automation for only one product type. Finally, the volume of goods produced is directly tied to the number of customer orders.

Project processes

Project processes are used to make one-at-a-time products exactly to customer specifications. These processes are used when there is high customization and low product volume, because each product is different. Examples can be seen in construction, shipbuilding, medical procedures, creation of artwork, custom tailoring, and interior design. With project processes the customer is usually involved in deciding on the design of the product. The artistic baker you hired to bake a wedding cake to your specifications uses a project process.

Batch processes

Batch processes are used to produce small quantities of products in groups or batches based on customer orders or product specifications. The volumes of each product produced are still small and there can still be a high degree of customization. The classes you are taking at the university use a batch process.

Note that both project and batch processes have low product volumes and offer customization. The difference is in the volume and degree of customization. Project processes are more extreme cases of intermittent operations compared to batch processes.

2. Continuous Operations (product-focused)

Continuous operations are used to produce one or a few standardized products in high volume. Resources are organized in a line flow to efficiently accommodate production of the product. Note that in this

environment it is possible to arrange resources in a line because there is only one type of product. This is directly the opposite of what we find with intermittent operations.

To efficiently produce a large volume of one type of product these operations tend to be *capitalintensive* rather than labor intensive. An example is “mass production” operations, which usually have much invested in their facilities and equipment to provide a high degree of product consistency. Often these facilities rely on automation and technology to improve efficiency and increase output rather than on labor skill. The volume produced is usually based on a forecast of future demands rather than on direct customer orders.

Line processes

Line processes are designed to produce a large volume of a standardized product for mass production. With line processes the product that is produced is made in high volume with little or no customization.

Continuous processes

Continuous processes operate continually to produce a very high volume of a fully standardized product. The products produced by continuous processes are usually in continual rather than discrete units, such as liquid or gas. Also, these facilities are usually highly capital intensive and automated.

Note that both line and continuous processes primarily produce large volumes of standardized products. Again, the difference is in the volume and degree of standardization. Continuous processes are more extreme cases of high volume and product standardization than are line processes.

The most common differences between intermittent and continuous operations relate to two dimensions:

- (1) the amount of product volume produced, and
- (2) the degree of product standardization.

Product volume can range from making a single unique product one at a time to producing a large number of products at the same time. *Product standardization* refers to a lack of variety in a particular product. The type of operation used, including equipment and labor, is quite different if a company produces one product at a time to customer specifications instead of mass production of one standardized product.

Figure 3-1 Types of processes based on product volume and product standardization

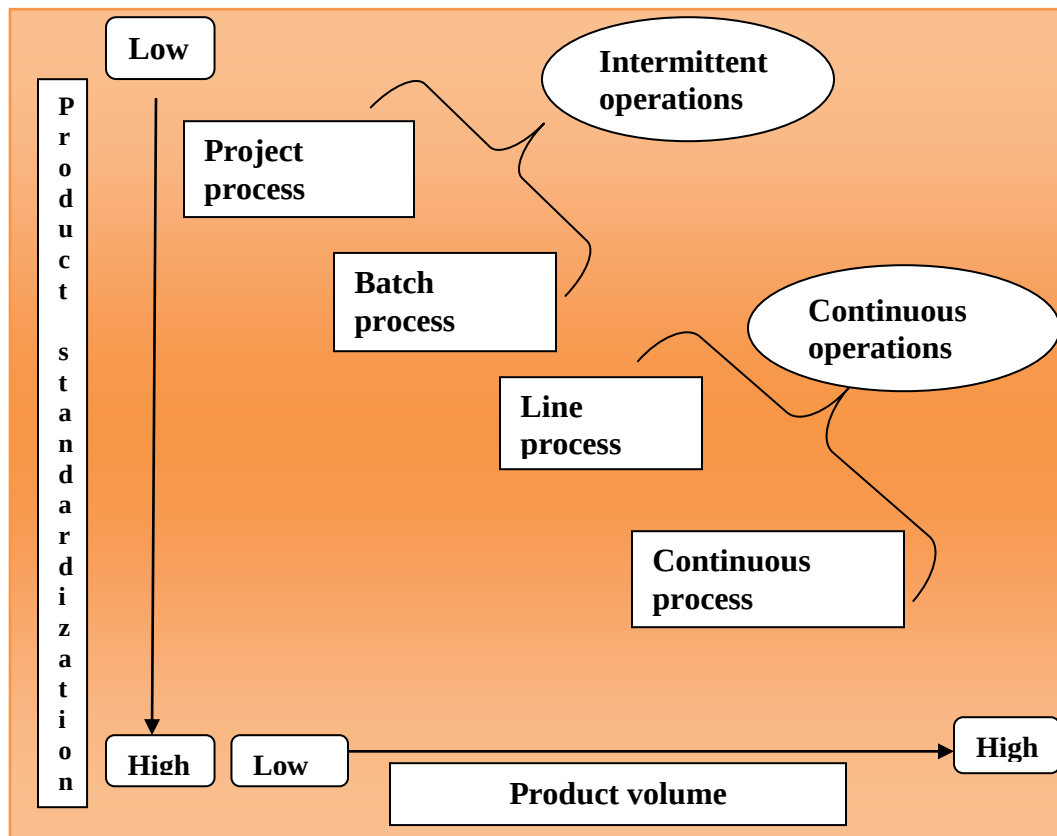


Figure 3-1 positions these four process types along the diagonal to show the best process strategies relative to product volume and product customization. Companies whose process strategies do not fall along this diagonal may not have made the best process decisions. Bear in mind, however, that not all companies fit into only one of these categories: a company may use both batch and project processing to good advantage.

CLASSIFICATION BY TYPE OF CUSTOMER ORDER

Another critical process choice is whether the product is make-to-order or make-to-stock. There are advantages and disadvantages to each type of process.

1. Make-to-stock /MTS/

It is a process that produced standard products which are stored in finished goods inventory. The product is delivered quickly to the customer from the finished goods inventory. MTS process can provide faster service to customer from available stock and lower costs than a make-to-order process. The MTS process is building products for inventory, and the jobs in process are not identified for any particular customer. The MTS process has a standard product line specified by the producer, not by the customer. The products are carried in inventory to immediately fulfill customer demand. Everything in operations is keyed to producing inventory in advance of actual demand in order to have the proper products in stock when customer calls. The critical management tasks are forecasting, inventory management, and capacity planning.

The MTS process begins with the producer specifying the product. The customer then requests a product from inventory. If it is not available, a back order may be placed or the order can be lost to the firm. In an MTS process customer orders cannot be identified during production. Performance measures for a MTS process include the percentage of orders filled from inventory. Other measurements are the length of time that it takes to replenish inventory, inventory turnover, capacity utilization and the time it takes to fill a back order. The objective of MTS process is to meet the desired service level at minimum cost.

In summary, the MTS process is keyed to replenishment of inventory with order fulfillment from inventory and its process is measured by service level efficiency in replenishing inventory.

2. Maker-to-order/MTO/

The process is activated only in response to an actual order. Inventory (both work in process and finished goods) is kept a minimum or no inventory. In the MTO order, individual orders can be identified during production. As each order is made to the customer specification, the jobs in process are actually associated with customers. The MTO process can have a wide range of order specification and has higher flexibility for product customization.

In the MTO process, the cycle of production and order fulfillment begins with the customer order. The key performance measures of a MTO process are the length of time it takes to design, make, and deliver the product. This is often referred to as lead time. Another measure of performance in MTO environment is the percentage of orders completed on time. This percentage can be based on the delivery date the customer originally requested or the date that was subsequently promised to the customer.

Summary, the MTO process is keyed to customer orders. An MTO process can provide higher level of

product variety and has greater flexibility for product customization. The MTO process is measured by its response time to customers and the efficiency in meeting its customer orders.

Summary: Make-to-stock Vs make-to-order

Characteristics	Make-to-stock	Make-to-order
Product	Product specified	Customer-specified
	Low variety	High variety
	Inexpensive	Expensive
Objectives	Balance inventory, capacity and service	Manage delivery lead times and capacity
Main operations problems	Forecasting	Delivery promises
	Planning production	Delivery time
	Control of inventory	

3.3 STRATEGIC CAPACITY PLANNING

How many units of equipment do we need to achieve our production forecast? This is the concept of capacity planning. Capacity can be defined as ability to produce certain output within a specified time period or the rate of output that can be achieved from a process. The word capacity normally defined in Business dictionary as “specific ability of an entity (person or organization) or resource, measured in quantity and level of quality, over an extended period” (BusinessDictionary.com, 2011).

Capacity also refers to the limitation which the operating element is able to process; the amount of services executed or tangible products produced. The vital elements and considerations needed to be taken into account before-hand are what type of capacity – whether it’s equipment, space or human skills – are needed, how much of it is required and the timeframe of when those factors are to be accessible.

Strategic capacity planning is an approach for determining the overall capacity level of capital intensive resources, including facilities, equipment, and overall labor force size.

Importance of Capacity Decisions

For a number of reasons, capacity decisions are among the most fundamental of all design decisions that managers must make.

- 1) Capacity decisions have a real impact on the ability of the organizations to meet future demands for products and services; capacity essentially limits the rate of output possible.

- 2) Capacity Decisions Affect Operating cost: Ideally, capacity and demand requirements will be matched, which will tend to minimize operating costs. In practice, this is not always achieved because actual demand either differs from expected demand or tends to vary (e.g. cyclically). In such cases, a decision might be made to attempt to balance the costs of over and under capacity.
- 3) Capacity is usually a major determinant of initial Cost. Typically, the greater the capacity of a productive unit, the greater its cost. This does not necessarily imply a one for-one relationship; larger units tend to cost *proportionately* less than smaller units.
- 4) Capacity decisions are often involve long-term commitment of resources and the fact that, once they are implemented, it may be difficult or impossible to modify those decisions without incurring major cost.
- 5) Capacity decisions can affect competitiveness: If a firm has excess capacity, or can quickly add capacity, that fact may serve as barrier against entry by other firms.

Factors affecting capacity

Capacity is affected by both internal and external factors.

The external factor includes: government regulations (e.g. working hours, safety, pollution etc), union agreement, and suppliers capabilities.

The internal factor includes: product and service design, personnel and jobs(worker training, motivation learning job content and methods), plant lay out and process flow, equipment capabilities and maintenance, materials management, quality control system, product mix decision, and management capabilities.

Capacity Expansion timing strategies

How much to increase or decrease capacity and when, is a strategic choice. For manufacturing firms, there are three major strategies for adding capacity: proactive, neutral and reactive. Each has its weaknesses and strengths. Which strategy to adopt is dependent, to a large extent, on the operating characteristics of the facility and the overall strategy of the firm?

a) Capacity lead strategy

A firm employs a capacity lead strategy when it intentionally invests capacity in advance of demand to eliminate the chance of losing sales to competitors. The economic tradeoff requires incremental profits from making those sales to exceed the incremental costs of operating below full capacity. As an alternative, they may value its current customers so highly that it invests in the extra capacity despite the

cost to protect its customer-service reputation.

b) Capacity lag strategy

A Capacity lag strategy calls for expansion investments only after confirmation of rises in demand in order to maintain high utilization rate. If a plant produces either a homogenous commodity or a standard product that appeals to the customer based primarily on cost, then this strategy will maximize profits by minimizing operating cost.

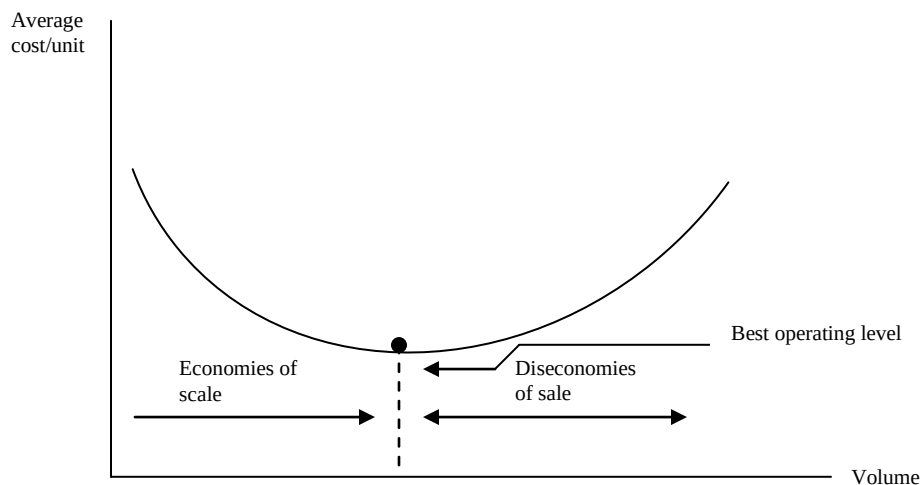
c) Capacity straddle strategy

A Capacity straddle strategy tries to keep abreast of growing demand by matching average capacity to average demand. It calls for expansion only when managers expect that they can sell at least some of the additional output, but before they know that they can sell it all.

Important Concepts of Capacity Decisions

☀ *Best operating level*

The best operating level is the level of capacity for which the average unit cost is at a minimum. Note that as we move down the unit cost curve for each plant size, we achieve economies of scale until we reach the best operating level and then diseconomies of scale as we exceed this point. This level of operation is shown in the figure below:



As we move down the curve, we achieve economies of scale until we reach the best operating level and we encounter diseconomies of scale as we exceed this point. The upward swing of unit cost as volume increases results from:

- using less efficient machines
- working overtimes

- increasing the cost of maintenance or
- using inexperienced or less skilled employees

☀ **Economics and diseconomies of scale**

The basic notion is well known: as a plant gets larger and volume increases, the average per unit of output drops because each succeeding unit absorbs parts of the fixed costs. Economics of scale is a concept which state that the average unit cost of goods or services can be reduced by increasing its output rate. There are four principal *reasons* for why economics of scale can drive cost down when output increases:

- ✓ *Fixed costs are spreads over more units:* the fixed cost includes heating cost, debt services, and management salaries. Depreciation of plant and equipment already owned is also a fixed cost in the accounting sense. When the output rate increases, the average unit cost drops because fixed costs are spread over more units.
- ✓ *Construction costs are reduced:* certain activities and expenses are required in building small and large facilities alike: building permits, architects' fees, rental of building equipment, and the like. Industries such as breweries and oil refineries benefits from strong economics of scale because of this phenomenon.
- ✓ *Costs of purchased materials are cut:* higher volume can reduce the cost of purchased materials and services. They give a purchaser a better bargaining position and the opportunity to take advantage of quantity discounts.
- ✓ *Process advantages are found:* high volume production provides many opportunities for cost reduction. At a higher output rate, the process shifts towards a line process, with resources dedicated to individual products. The benefits from dedicating resources to individual products or services may includes spreading up the learning effects, lowering inventory, improving process and job design, and reducing the number of changeovers.

☀ **Diseconomies of scale**

At some point a facility can become so large that diseconomies of scale set in; that is, the average cost per unit increases as the facility size increase the reason is that excessive size can bring complexity, loss of focus, and inefficiencies that raise the average unit cost of a product or services. There may be too many layers of employees and bureaucracy, and management loses touch with employees and customers. The organization is less agile and loses the flexibility needed to respond to changing demand. Many large

companies become so involved in analysis and planning that they innovate less and avoid risks. The result is that small companies outperform corporate giants in numerous industries.

☀ *Learning (experience) curve*

Learning (experience) curve theory has a wide range of application in the business world. In manufacturing, it can be used to estimate the capacity requirement and the time for product design. Learning curves can be applied to individuals or organizations. Individual learning is improvement that results when people repeat a process and gain skill or efficiency from their own experience. That is 'practices make perfect'. Organizational learning results from practices as well, but it will also come from changes in administration, equipment, and product design. In organizational settings, we expect to see both kinds of learning occurring simultaneously and often describe the combined effect with the single learning curve.

Generally, it is quite possible that initially the operator takes longer time to accomplish the job as compared to the subsequent cycles when he would have acquired the necessary skill and feel in 'learning' the job. Usually, this learning curve is hyperbolic in nature. Though the learning curve concept is important one, it has not been given due consideration. Scholars feel it would be unfair if learning phase is not accounted for while determine capacity requirement and time standard.

☀ *Capacity focus*

The concept of the focused factory holds that production facilities work best when they focus on a fairly limited set of production objectives. This means that a firm should not expect to excel in every aspect of manufacturing performance: cost, quality, flexibility, short lead time, and low investment. Rather, it should select a limited set of tasks that contribute the most to corporate objectives.

☀ *Capacity bottlenecks*

It is an operation that has the lowest effective capacity of any operation in the process and thus limits the systems output. True expansion of a process's capacity occurs only when bottleneck capacity is increased. The long term capacity bottlenecks can be expanded in various ways. Investment can be made in new equipment; bottleneck's capacity can also be expanded by operating it more hours per week such as going from one shift operation to multiple shifts or going from five work days per week to seven work days per week. Managers also might relieve the bottle neck by redesigning the process.

Long term capacity expansion is not the only way to ease bottlenecks. Overtime, temporary or part-time employees, or temporarily outsourcing or sharing during peak demand period are short term options.

☀ **Capacity flexibility**

Capacity flexibility means having the ability to rapidly increase or decrease production levels, or to shift production capacity quickly from one product or service to another. Such flexibility is achieved through:

- ✓ *Flexible plants*: perhaps the ultimate plant flexibility is the zero-changeover time plant. Using movable equipment, knockdown walls, and easily accessible and re routable utilities e.g. tents. Such a plant can adapt to change in real time.
- ✓ *Flexible process*: flexible processes are epitomized by flexible manufacturing systems on the one hand and simple, easily set up equipment on the other hand. Both of these technological approaches permit rapid low cost switching from one product line to the other, enabling what is referred to as *economics of scope*. By definition, economics of scope exist when; multiple products can be produced at a lower cost in combination than they can separately.
- ✓ *Flexible workers*: flexible workers have multiple skills and the ability to switch easily from one kind of task to another. They required broader training than specialized workers and need managers and staff support to facilitate quick changes in their work assignment.

Measures of capacity

No single capacity measure is applicable to all types of situations. For example, a retailer measure capacity as annual sale dollars generated per square foot, a theater measure capacity as number of seats, and a job shop measure capacity as number of machine hour.

Important terms used to measure the capacity are:

A. Design Capacity

Design capacity (peak capacity) is the maximum rate of output achieved under ideal condition. Design capacity values are stated by the manufacturer of the equipment. It may and commonly does include recognition of the need for routine maintenance but does not include recognition of delays caused by factors like scheduling, conflicts, defective products, low quality material, or change in product mix. In other words, manufacturers cannot anticipate the actual conditions of use. Therefore, this level of capacity cannot to be achieved under the real situation.

B. Effective Capacity

Effective capacity is the maximum output that a process or firm can economically sustain under normal conditions. It is the greatest level of output the firm can reasonably sustain by using realistic employee work schedules and the equipment currently in place.

It is the maximum possible given predicted problems such as a product mix, problems in scheduling and balancing operations, machine maintenance, quality factors, and so on. It also includes lunch breaks, and coffee breaks. It is typically less than or equal to the design capacity. In another word effective capacity is rarely equivalent to design capacity and is frequently much lower.

C. Actual or Operating Capacity

Actual capacity is the actual output of a system at a given point in time. In other words, it is the rate of output actually achieved. That is, the actual output produced in a real condition. It is even less than effective capacity, for it is affected by unpredicted short range factors such as equipment break down, absenteeism, shortage of raw materials, productivity, and other factors that are outside the control of the operations manager.

These different measures of capacity are useful in defining two measures of system effectiveness: *efficiency* and *utilization*.

- ✓ **Capacity utilization:** is the degree to which equipment, space or labor is currently being used. It the ratio of capacity used during a fixed period of time to the available capacity during that same time period.

Utilization is a measure relating design capacity to output. It is calculated as follows:

$$Utilization = \frac{\text{actual capacity (or capacity used)}}{\text{Designed capacity}} \times 100\%$$

- ✓ **Efficiency** is a measure of the use of effective capacity in producing a particular result.

$$Efficiency = \frac{\text{actual out put}}{\text{Effective capacity}} \times 100\%$$

- ✓ **Rated capacity:** when capacity is measured relative to equipment alone, the appropriate measure is rated capacity. It is an engineering assignment of maximum annual output, assuming continuous operations except for an allowance for Norman maintenance, and repair downtime.

$$Rated\ capacity = design\ capacity \times effective\ capacity \times efficiency$$

Rated capacity will always be less than or equal to effective capacity.

$$Design\ (peak)\ capacity > effective\ capacity > rated\ capacity$$

Example 1

If operated around the clock under ideal conditions, the fabrication department of an engine manufacturer can make 100 engines per day. Management believes that a maximum output rate of only 45 engines per day can be sustained economically over a long period of time. Currently, the department is producing 50 engines per day. What is the utilization of the department related to designed capacity? And compute Efficiency?

Solutions

Given: *actual output* = 50, *effective capacity* = 45, and *designed capacity* = 100 engine

$$Utilization = 50/100 \times 100\% = 50\%$$

$$Efficiency = 50/45 \times 100\% = 111\%$$

Example 2

The Sara James Bakery has a plant for processing breakfast rolls. The facility has an efficiency of 90%, and the effective capacity is 80%. Three process lines are used to produce the rolls. The line operate 7 days a week and three 8 hours shift per day. Each line was designed to process 120 standard rolls per hour. What is designed capacity? Rated capacity?

Solution

$$Number\ of\ lines = 3,$$

$$number\ of\ hours = 7\ days \times 24\ hrs\ per\ day = 168\ hrs$$

$$number\ of\ rolls/hrs = 120$$

$$\begin{aligned} Designed\ capacity &= number\ of\ lines \times number\ of\ hours \times number\ of\ roles\ per\ hour \\ &= 3 \times 168 \times 120 \end{aligned}$$

$$= 60,480\ rolls\ per\ week.$$

$$Rated\ capacity = 60480\ rolls\ per\ week \times 0.8 \times 0.9$$

$$= 43546\ rolls\ per\ week$$

CAPACITY PLANNING DECISION

Capacity planning is central to long-term success of an organization. Too much capacity can be as agonizing as too little capacity. The objective of capacity planning is to specify which level of capacity will meet market demand in cost efficient way. Capacity planning is generally viewed in three time duration:

Long range (greater than one year):- when productive resources take a long time to acquire or dispose of. Example: building, equipment or facilities decisions. Long range planning requires top management participation and approval.

Intermediate range: - monthly or quarterly plans for the next 6 to 18 months. Here capacity may be varied by such alternatives as hiring part timer, layoff, minor equipment purchase and sub contracting.

Short range: - less than one month. This is tied into the daily or weekly scheduling process and involves making adjustment to eliminate the variance between planned output and actual output. This includes alternatives such as overtime, personnel transfers, and alternative production routings.

A Systematic Approach to Capacity Decisions

Although each situation somewhat different, a four step procedure generally can help managers make sound capacity decisions. In describing this procedure, we assume that management has already performed the preliminary step of determining existing capacity.

Step 1: Estimate future capacity requirements

The foundation for estimating long term capacity needs is forecasts of demand, productivity, competition, and technological changes that extend well into the future. Unfortunately, the farther ahead you look the more chance you have of making an inaccurate forecast.

The demand forecast has to be converted to a number that can be compared directly with the capacity measure directly used. Suppose that capacity is expressed as the number of available machines at an operation. When just, one product (services) is being processed, the number of machines required, M, is (for single product).

☞ One type of product

No. of machines required

$$= \frac{\text{Processing hours required for year's demand}}{\text{Hours available for one machine per year, after the desired cushion deducted}}$$

$$M = \frac{Dp}{N [1-(c/100)]}$$

Where, D = No. of unit (customer) forecast per year

p = processing time (in hours per unit or customer)

N = total number of hours per year during which the process operate

C = desired capacity cushion

☞ More than one type of product: n types of products

If multiple products or services are involved, extra time is needed to change over from one product to the next. Set up time is the time required to change a machine from making one product or service to making another.

Total setup time = $D \div$ (by the average lot size)

When there are multiple products (services)

No. of machines = processing and set up hours required for year's demand,

Required $\frac{\text{Summed over all products}}{\text{Hrs available for one machine per year, after deducting desired cushion}}$

$$M = \frac{\left[Dp + \left(\frac{D}{Q} \right) S \right] \text{product 1} + \left[Dp + \left(\frac{D}{Q} \right) S \right] \text{product 2} + \dots + \left[Dp + \left(\frac{D}{Q} \right) S \right] \text{product } n}{N(1 - C)}$$

Where, Q = number of units in each lot

S = set up time (in hours) per lot

Note: Always round up the fractional part for the number of machines required.

Example: a copy center in an office building prepares bound reports for two clients. The center makes multiple copies (the lot size) of each report. The processing time to run and bind each copy depends on the number of pages. The center operates 250 days per year, with one eight –hours shift. Management believes that a capacity cushion of 15% is best. If currently have three copy machines. Based on the following table of information, determine how many machines are needed at the copy center.

Item	client X	client Y
Annual demand (copies)	2,000	6000
Standard processing time (hrs/copy)	0.5	0.7
Average lot size (copies per report)	20	30
Standard setup time (hours)	0.25	0.40

Solution

$M = \frac{[(2000 \times 0.5) + (2000/20) \times 0.25] + [(6000 \times 0.7) + (6000/30) \times 0.4]}{[(250 \text{ days/year}) (1 \text{ shift/day}) (8 \text{ hrs/shift})] (1.0 - 15/100)} = 5,305/1700 = 3.12$ (Round up to the next integer gives a requirement of 4 machines).

Step2: Identify gaps

A capacity gap is any differences (positive or negatives) between projected demand and current capacity. Identifying gaps requires use of the correct capacity measures. Complications arise when multiple operations and several resource inputs are involved. Expanding the capacity of some operations may increase over all capacity. However if one operation is a bottle neck, capacity can be expanded only if the capacity of the bottlenecked operation is expanded. Bottle neck is any resource whose capacity is less than the demand placed up on it.

Example: grandmother's chicken restaurant is experiencing a boom in business. The owner expects to serve a total of 80,000 meals this year. Although the chicken is operating at 100% capacity, the dining room can handle a total of 105,000 diners per year. Based on the following forecasted demand for the next 5 years, determine the capacity gap.

Year	Meals	Capacity Gap	Demand Capacity
1	90,000	90,000 - 80,000 = 10,000	capacity is positive
2	100,000	100,000 - 80,000 = 20,000	capacity is positive
3	110,000	110,000 - 80,000 = 30,000	110,000 - 105,000 = 5,000
4	120,000	120,000 - 80,000 = 40,000	120,000 - 105,000 = 15,000
5	130,000	130,000 - 80,000 = 50,000	130,000 - 105,000 = 25,000

NB:
The
kitchen
is
currentl
y the
bottlene

ck at 80,000 meals per years.

For the first two years the capacity of dining room (105,000) is greater than demand. Capacity gaps exist for the last three years as shown above.

Step3: Develop alternatives

After determining capacity gap, the next logical step is to develop alternative plans to cope with projected gaps. One alternative, called the **base case**, is to do nothing and simply lose orders from any demand that exceeds current capacity. Other alternative are various timing and sizing options for adding new capacity. Additional possibilities includes: expanding at a different locations and using short term options such as overtime, temporary workers, and subcontracting.

Step4: Evaluate the alternatives

In this step, the managers evaluate each alternative, both quantitatively and qualitatively.

Qualitative concerns: qualitatively the manager has to look at how each alternative fits the overall capacity strategy and other aspects of the business not covered by the financial analysis. Of particular concerns might be uncertainties about demand, competitive reaction, technological change, and cost estimate. Some of these factors cannot be quantified and has to be assessed on the bases of judgment and experience. Others can be quantified, and the managers can analyze each alternative by using different assumptions about the future. One set of assumption could represent a worst case, where demand is less, competition is greater, and construction costs are higher than expected. Another set of assumption could represent the most optimistic view of the future. These types of what-if analysis allow the manager to get an idea of each alternative's implications before making a final choice.

Quantitative concerns: quantitatively, the manager estimates the changes in cash flows for each alternative over the forecasted time horizon compared to the base case. Cash flow is the different between the flows of funds in to and out of an organizations over a period of time, including revenues, costs, and changes in assets and liabilities.

Step5: Finally, based on the evaluation results, the manager must make the choice

Selecting a capacity alternative which is the most suited to achieve strategic mission.

Following these steps, organization should design the right capacity, that is, the capacity best matches with the demands of the product. However, there are several reasons why the production capacity to be provided does not necessarily equal the amount of products and services expected to be demanded. First enough capital and other resources may not be economically available to satisfy all of the demand. Secondly, because of the uncertainty of forecasts and the need to link production capacity to operations strategy interns of competitive priorities, a capacity cushion may be provided. A capacity cushion is an additional amount of production capacity added onto the expected demand to allow;

1. Extra capacity incase of more demand than expected occurs
2. The ability to satisfy demand during peak demand seasons.
3. Lower production costs; production facilities operated to close to capacity experience higher costs.
4. Product and volume flexibility responding to customers' needs for different products and high volumes is possible because of the extra capacity.

Ways of changing long range capacity

A. Expansion

Expansion would take either or a combination of the ways;

- Sub-contract with other companies to become suppliers of the expanding firm's components or entire products.
- Acquire other companies, facilities or resources
- Develop sites, build buildings, by equipment
- Expand, update, or modify existing facilities or
- Reactivate facilities on standby status.

B. Reduction

This strategy requires managers to take the following actions when expansion is not appropriate due low demand or any other internal and external factors.

- Sell of existing facilities; sell inventories, and layoff or transfer employees.
- Mothball facilities and standby status, sell inventories, and layoff or transfer employees.
- Develop and produce new products as other products decline.

TOOLS FOR CAPACITY PLANNING

An organization needs to examine alternatives for future capacity from a number of different perspectives. Most obvious are economic considerations. Such as; will an alternative be economically feasible? How much will it cost? How soon can we have it? What will the operating and maintenance costs be? What will its useful life be? Will it be compatible with present personnel and present operations?

Long term capacity planning requires demand forecasts for an extended period of time. Unfortunately, forecast accuracy declines as the forecasting horizon lengthens. In addition, in anticipating what competitors will do increases the uncertainty of demand forecasts. Finally, demand during any period of time is not evenly distributed; peaks and valleys of demand may (and often do) occur within the time period. These realities necessitate the use of capacity cushion.

A number of techniques are useful for evaluating capacity alternatives from an economic standpoint. Some of the more common are cost-volume analysis (Break-even analysis), financial analysis, decision theory, and waiting line analysis. In this section, only cost volume analysis (break-even analysis), waiting line and decision tree are discussed.

1. Break-Even Analysis

Though different tactics can be used to adjust demand to existing facilities, the strategic issue is, of course, how to have facility of the correct size. Break-even analysis may help with that decision.

Breakeven can aid capacity decisions by identifying the processes with the lowest total cost for the volume expected. The objective of break-even analysis is to find the point, in dollars and units, at which costs equal revenues-which is the break-even point. Break-even analysis requires an estimation of *fixed costs*, *variable cost*, and *revenue*.

Fixed costs are costs that continue even if no units are produced such as depreciation, taxes, debt and mortgage payments where as variable costs are those that vary with the volume of units produced. The major components of variable costs are labour and materials and other costs such as the portion of the utilities that varies with volume.

Another element in break-even analysis is the revenue function that begins at the origin and proceeds upward to the right increasing by the selling price of each unit. Where the revenue function crosses the total cost line is the break-even point, with a profit corridor to the right and a loss corridor to the left. Break-even analysis assumes that costs and revenue increase in direct proportion to the volume of units being produced. However, neither fixed costs nor variable costs (nor, for that matter, the revenue function) need be a straight line.

To utilize the concept of breakeven analysis for capacity planning decision, we first define our goal such as a profit level, and then work back to determine the size of facility to be owned so that its production capacity can effectively lead to the production level required (i.e., quantity) to achieve a goal

Example:

XYZ Company is now contemplating to adding new line of product, which requires leasing new machine for a monthly payment of Br. 6000. Variable costs would be Br.2.00 per unit, and the product would be sold for Br.7.00 each.

Required:

- a. What should be the monthly production capacity of a machine for achieving a breakeven point?
- b. What should be a production capacity so that a firm can achieve a profit target of Br. 4000?

2. *Waiting line models*

Waiting line models often are useful in capacity planning. Waiting line tend to develop in front of a work centre, such as an airport ticket counter, a machine centre, or a central computer. The reason is that the arrival time between jobs or customers vary and the processing time may vary from one customer to the next. Waiting line model use probability distribution to provide estimate average customer delay time, average length of waiting lines, and utilization of the work centre. Managers can use this information to choose the most cost effective capacity, balancing customer services and the cost of adding capacity.

3. *Decision trees*

Decision tree is a tree like diagram that depicts alternatives and their possible outcomes. This tool can be used to evaluate alternative capacities and enable managers make appropriate decisions. A decision tree can be particularly valuable for evaluating different capacity expansion alternatives when demand is uncertain and sequential decisions are involved. A decision tree is a systematic model of the sequence of steps in a problem and the conditions and consequences of each step.

Example

A firm that plans to expand its product line must decide whether to build a small or a large facility to produce the new products. If it builds a small facility and demand is low, the net present value (NPV) after deducting for building costs will be Br. 400,000. If demand is high, the firm can either maintain the small facility or expand it. Expansion would have a net present value of Br. 450,000 and maintaining the small facility would have a net present value of Br. 50,000. If a large facility is build and demand is high, the estimated NPV is Br. 800,000. If demand turns out to below, the NPV will be Br. 10,000. The problem that demand will be high is estimated to be 0.60, and the problem of low demand is estimated to be 0.4.

Required:

- a. Draw the tree diagram.
- b. Which alternative capacity should be build? What is the expected NPV of the alternative chosen?

3.4 FACILITY LOCATION & LAYOUT

3.4.1 FACILITY LOCATION

The choice of location for business organization is an important issue in the design of the production system. Where should a plant or service facility be located? This is a top question on the strategic agendas

of contemporary manufacturing and service firms, particularly in this age of global markets and global production. Globalization allows companies greater flexibility in their location choices. However, in practice, the question of location is very much linked to two competitive imperatives.

1. The need to produce close to the customer due to time based competition and shipment costs.
2. The need to locate near the appropriate resource pool to take advantage of low costs.

Location decision is an integral part of the strategic planning process of every organization. Although it might appear that location decision are mostly one-time problem pertaining to new organization, the fact is that existing organization often have a bigger stake in these kinds of decisions than new organization. In other words, location problems are common to new and existing businesses.

Facility location decisions are, strategic, long term and non repetitive in nature. Without sound and careful location planning in the beginning itself, the new facility may pose continuous operating disadvantages, for the future operations. Location decisions are based on a host of factors; some are subjective, qualitative, and intangible while some others are objectives, quantitative and tangible.

The Need for location Decisions

Existing organizations become involved in location decision for a variety of reasons. The following are some of the reasons for such decisions (other than the need for greater capacity).

1. Opportunity for expanding market share

From such as banks, fast food chains, supermarkets, and retail stores view location as part of marketing strategy, and they look for locations that will help them to expand their markets. Basically, the location decisions in those cases reflect additional new location to existing suppliers.

2. Business growth in demand

A similar situation occurs when an organization experiences a growth in demand for its products or service that cannot be satisfied by expansion at an existing location. The addition of a new location to complement an existing system is often a realistic alternative.

3. Depletion of Basic resources

Some firms become presented with location decision because of the depletion of basic inputs. For example fishing and logging operations are forced to relocate due to the temporarily exhaustions of fish or forest at a given location. Mining and petroleum organization face the same sort of situation, although usually over a longer time horizon.

4. Shift in Market /demand

If the demand for the product does not exist in the existing location, it is a good reason to consider and find out a better location.

5. Operating Costs

Cost of doing business in a particular location reaches a point where other locations begin to look more attractive. In this case, the company may shift to a cost effective location.

6. Merge of companies

Merger of companies changes the ownership titles and may require change in management and operation of the merging firms, and then leading to location decisions.

7. Introduction of new product.

This may require to a new resource, labour or material which may not exist in the existing location. Therefore, firms make a location decision to produce a sell their new product.

Characteristics of location decision

1. Location decisions entail a long-term commitment, which makes mistakes difficult to overcome. In addition, location decision often has an impact on operating costs both fixed and variables and revenues as well as an operation. Example, a poor choice of location might result in excessive transportation cost, shortage of qualified labour, loss of competitive advantage, shortage of raw materials and location of customer (operation problem).
2. Location decision requires the selection of location from a number of acceptable location instead of identifying the “One best” location. If one site is clearly superior to all others in all respects, the location decision is an easy one. However, several site candidates, each with its strengths and weaknesses emerge as good choice and the location decision becomes a trade off decision.
3. Location decision involves four *options* that managers can consider in location planning. These are:
 - A. **Expanding an existing facility:** These options can be attractive if there is adequate room for expansion, especially if the location has desirable features that are not readily available elsewhere. Expansion costs are often less than those of other alternatives.
 - B. **Adding new location:** Another option is to add new location while retaining existing ones, as it is done in many retail operations. The advantage of this option are: it draws /attracts customers who are already looking for an existing business, and used as a defensive strategy designed to maintain a market share or prevent competitors from entering a market.
 - C. **Shutting down:** The third option is to shut down at one location and move to another. An organization must weigh the cost of a move and the resulting benefits against the costs and benefits

and remaining in an existing location. This option is considered when market shifts, exhaustion of raw materials and the cost of operation often cause firms to seriously consider this option.

- D. **Doing nothing:** If a detailed analysis of potential locations fails to uncover benefits that make one of the previous three alternatives attractive, a firm may decide to maintain a status of at least for the time being.

The Location Decision Hierarchy and Factors that Affect Location Decision

There are four location decision hierarchies:

i. Global- international considerations

It is the highest level in the location decision hierarchy. Decision makers who are considering expanding in to a new country must consider macroeconomic, demographic, and political issues of long term significance. They must consider international trade issues, such as

- International trade issues (currency exchange risk, balance of trade, quotas, tariffs etc.)
- Market access issues (such as free trade agreement, consumer sentiment towards imported goods)
- Labor issues (availability, wages, skill and training, and regulations)
- Political concerns (stability of current regime, risk of asset nationalization, local owner ship laws etc.)
- Cultural issues (compatibility of business practices and products with local culture)
- Legal issues(environmental regulations, accounting &reporting requirements etc)

ii. Regional considerations

Once the decision has been made to locate a facility in a particular country, decision makers must choose a region based on regional issues like:

- *Supply issues(availability of material inputs):* select the region endowed with abundant or sufficient inputs.
- *proximity to market:* select the region which is near to the major market if product is perishable, fragile, if the product needs large transportation space, transportation cost is expensive,and if further processing increases the volume, bulk, fragility and perish ability.
- *proximity to material:* a possible reason to select regions which is near resources includes: when further processing reduces the bulk (e.g. sugar processing), perish ability (e.g., freezing, canning, pasteurizing etc)
- *Transportation facility:* transformation facilities are essential for the economic operations of production systems. Operations manager must study the characteristics of the new materials and

finished products in regard to their transportation (water, railroad, road, pipelines, and air line transport) need and search for the location (region) that provides facilities of transportation with a reasonable cost.

- *Business Climate*: select the region with favorable climate which is important in order to acquire and maintain productive work forces. Certain industries such as agricultural business require specific climatic conditions.

iii. Community considerations

Choice of community depends on the following factors:

- *Community attitudes*: in order to ensure the long term existences in that community, it is mandatory to win the interest, enthusiasms, and cooperation of the society. Otherwise, poor relation with community will result in putting the survival of the organization under question mark. So select the community with positive attitudes towards the company.
- *Community government and financial incentives*: it is important to assess the current situation and attempt to predict the future situations in regard to the policies of local government. In general stable, competent, honest, and cooperative government officials are great asset to a newly located company. Local government may be evaluated in terms of financial incentives they offer, taxation policies, peace and securities etc.
- *Community facilities*: is concerned with the availability of schools, churches, medical facilities, residential housings, recreational opportunities, highways etc.

iv. Site considerations

After identifying the community in the already selected region, the final step in location selection decision is to screen out the best site out of the possible ones. The followings are the basic factors that influence site selection.

- *Size and cost of the site*: while selecting exact site, consider the size of the site and its cost (building and construction costs). Generally, the size of the site must be large enough to satisfy requirements such as employees parking requirement & future expansion plans and the cost of the land must be reasonable.
- *Drainage and soil condition*: poor drainage leads to accumulations of water around the plant which may be harmful for some organization. Similarly, if the load bearing capacity of the soil is low, it will be difficult to establish sound building foundations.

- *Land development cost*: cost related to excavation, grading, filling, construction of road, sidings, etc must also be taken in to consideration while selecting the site.
- *Utilities*: selection of site is influenced by cost of acquiring and using utilities like electricity, natural gas, water as well as disposal facilities. For example all enterprises need safe and pure water. Some organizations like breweries need water even with some extra ordinary quality. So select the site which is better furnished with utilities.
- *Access concerns*: select the site which is easily accessible for customers, suppliers, utilities and other concerned bodies.

Methods of evaluating potential locations

We have seen that a variety of factors are important to decision makers at each level of the location decision hierarchy. What is their importance and which factor is most important? The answers to these questions vary from one decision to the next; there may be no precise answer. But, as shown below, there are some methods (both qualitative and quantitative) which are used to evaluate and compare potential site locations.

i. Factor rating method

It is a general approach that is useful for evaluating a given alternative and comparing alternatives. Rating the factors according to their general importance can help decision makers to avoid placing too much emphasis on the wrong factors. Multiple factor rating system can be used to compare the attractiveness of several locations on the basis of more than one criterion. Factor rating system is probably one of the most widely used location selection techniques because they can be combine very diverse issues in to an easy – to-understand format. At the same time, it is important to recognize the fact that although the end result from this type of analysis is a quantitative number, factor rating systems are used to evaluate both qualitative and quantitative factors. Another reason that the factor–rating system approach is so popular is that it is relatively simple to use, requiring only six steps.

- ✓ Identify the specific criteria/factors to be considered in selecting a site.
- ✓ Assign a weight to each factor indicating its importance relative to all of the other factors that are being considered. Typically weight sum to 1
- ✓ Select a common scale for rating each factor (for example 1-100)
- ✓ Rate (score) each potential location on each of the factors.
- ✓ Multiply each factor's score by the weight assigned to that factor

- ✓ Sum up the weighted scores for all of the factors and select that location with the highest total score.

Illustration

To illustrate the factor rating system, consider the following site selection problem. Three potential sites have been identified. Management has decided to use the following criteria and has assigned the following weights to each of them based up on their relative importance. The three locations are then rated on each of these factors and a total score for each location is calculated as follows:

Location rating 1 to 100.

Factor/criteria	Weight	Locations (max score 100)			Weighted score		
		A	B	C	A	B	C
1. Convenient	0.15	80	70	60	12.0	10.5	9
2. Parking facilities	0.20	72	76	92	14.4	15.2	18.4
3. Display area	0.18	88	90	90	15.84	16.2	16.2
4. Shopper traffic	0.27	94	86	80	25.38	23.22	21.6
5. Operating costs	0.10	98	90	82	9.8	9	8.2
6. Neighborhood	0.10	96	85	75	9.6	8.5	7.5
Sum	1.00				87.02	79.92	73.6

Decision: - Location A is better because it results in a highest score.

ii. Location break even analysis

This method is the use of cost-volume-profit analysis to make an economic comparison of location alternatives. By identifying fixed and variable costs and grouping them for each location, we can determine which one provides the lowest cost. Location break-even analysis can be done mathematically or graphically. The graphic approach has the advantage of providing the range of volume over which each location is preferable.

The three steps to the location breakeven analysis are:

- i. Determine the fixed cost and the variable cost associated with each location alternative.
- ii. Plot the total-cost line for all location alternatives on the same graph.
- iii. Select the location that has the lowest total cost for the expected level of output.

Assumptions of this method are:

- ✓ Fixed costs are constant for the range of probable output.
- ✓ Variable costs are linear for the range of probable output.
- ✓ The required level of output can be closely estimated.

- ✓ Only one product is involved.

Example

The location of a tractor factory in site **A** will result in certain annual fixed costs, variable costs and revenue. The figure would be different for site **B** and **C**. The quantity is 5,000 and the fixed and variable costs for three potential locations are given below. Which location is best?

Site	FC	variable cost/unit
A	10,000	60
B	30,000	40
C	90,000	20

Solution

To solve this problem, we can use graphical method or profit analysis method.

a) graphical method

Compute TC for two different output levels (say 0 units and 5000 units).

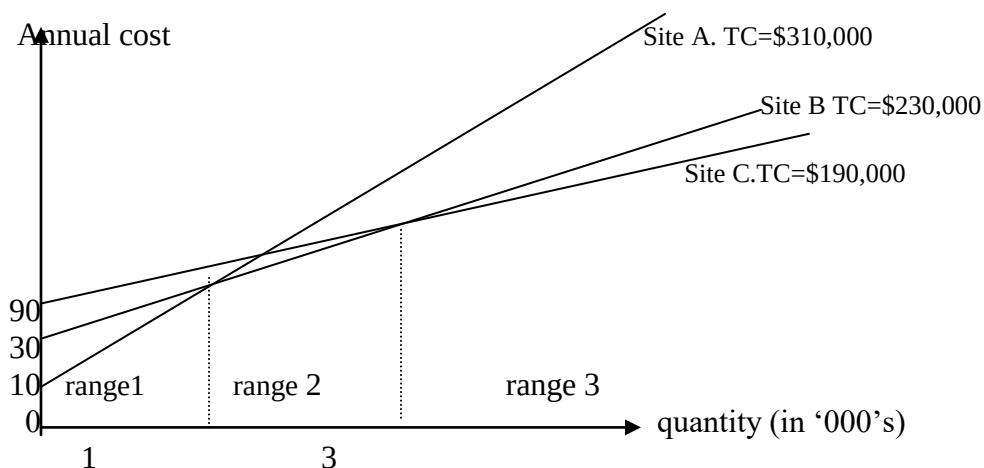
Case1. TC at zero output level

If $Q=0$, then $TC=FC$

Case2. TC (in \$) at 5000 output level

Site	FC	total variable cost	total cost (FC+TVC)
A	10,000	$60 \times 5000 = 300,000$	310,000
B	30,000	$40 \times 5000 = 200,000$	230,000
C	90,000	$20 \times 5000 = 100,000$	190,000

After calculating the TC for two different output levels, plot the cost curve for all the communities on a single graph.



Using break even analysis, determine the break even quantity. Break even quantity indicates the relevant range over which each site is feasible.

Range (break even quantity) over which each site provides lowest cost can be calculated as follows:

Range1. Equate the cost equation of site A&B, that is:

$$10,000 + 60Q = 30,000 + 40Q, \text{ then } Q = 1000 \text{ units}$$

Range 2. Equate the cost equation of site B & C that is:

$$30,000 + 40Q = 90,000 + 20Q, \text{ then } Q = 3000 \text{ units}$$

Range 3. At range 3, $Q > 3000$

Thus, if output is less than 1000 unit, location A is best. If output is between 1000 & 3000 units, site B is best. For output level greater than 3000 units, site C is best.

b) Profit analysis method

Given the above example, which site is best for expected demand of 2000 units if price per unit is 75, 68 & 80 for site A, B, & C respectively?

Solution

Calculate the total profit for each site at 2000 output level and select the site with the highest profit. Profit = TR - TC. Accordingly,

$$\text{Profit for site A} = (75 \times 2000) - [10,000 + (60 \times 2000)], \text{ Profit} = \$20,000$$

$$\text{Profit for site B} = (68 \times 2000) - [30,000 + (40 \times 2000)], \text{ then profit} = \$26,000$$

$$\text{Profit for site C} = (80 \times 2000) - [90,000 + (20 \times 2000)], \text{ then, profit} = \$30,000$$

Thus, site C is the best location.

iii. Centre of gravity method

This method is a quantitative method for determining the optimal site for a facility based up on minimizing total distribution cost. The first step in the centre of gravity method is to locate each of the existing retail operations on the X and Y coordinate grid map. The purpose of the grid map is to establish relative distance between the sites. The centre of gravity or the site for distribution facility is then found by calculating the X and Y coordinate that result in minimizing the distribution costs among all facilities. To determine the site the following formula can be used.

$$C_x = \frac{\sum d_{ix} V_i}{\sum V_i} \qquad C_y = \frac{\sum d_{iy} V_i}{\sum V_i}$$

Where, C_x is X coordinate of the centre of gravity

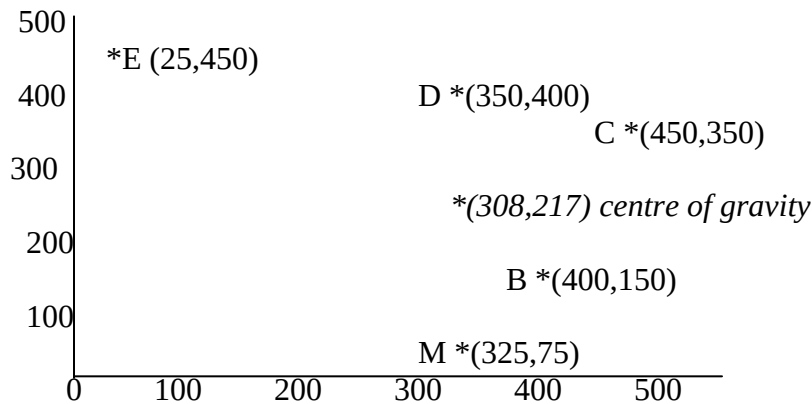
C_y = Y Coordinate of the centre of gravity

d_{ix} = X Coordinate of the i^{th} location

d_{iy} = Y Coordinates of the i^{th} location

V_i = volume of goods transported to the i^{th} location

Example: A refining company needs to locate an intermediates ware house facilities between its refinery plant in place M and its major distributors B, C, D & E. The following shows the coordinate map for both the plants and distributors.



More over shipping volume from plant M to major distributor are given as follows

Location	M	B	C	D	E
Gallon of gasoline Per month (000)	1500	250	450	350	450

Solution

Using the above formula, we can calculate the coordinates of the centre of gravity i.e. the site of the new location.

$$Cx = \frac{(325 \times 1,500) + (400 \times 250) + (450 \times 450) + (350 \times 350) + (25 \times 450)}{1,500 + 250 + 450 + 350 + 450} = 307.9$$

$$Cy = \frac{(75 \times 1,500) + (150 \times 250) + (350 \times 450) + (400 \times 350) + (450 \times 450)}{1,500 + 250 + 450 + 350 + 450} = 216.7$$

The location of the new facility should be at 308 and 217 i.e. (308,217)

3.4.2 FACILITY LAYOUT

Facility layout refers to an optimum arrangement of different facilities including man, machinery, equipment, materials, etc. within the factory building in such a manner so as to have quickest flow of material at the lowest cost and with the least amount of handling in processing the product from the receipt of material to the shipment of the finished product. Facility layout can also be defined as a decision

about the physical arrangement of anything that consumes spaces: person or group of people, a teller window, a machine, a work bench or work station, a department, storage room, a stair way or aisle and so on.

Since a layout once implemented it cannot be easily changed and costs of such changes are substantial, the facilities layout is a strategic decision. A poor layout will result in continuous losses in terms of higher efforts for material handling, more scrap and rework, poor space utilization etc. Hence, need to analyze and design a sound plant layout can hardly be overemphasized. It is a crucial function that has to be performed both at the time of initial design and of any facility, and during its growth, development and diversification.

The objective of facility layout is to allow workers and equipments to operate most effectively through appropriate arrangement of resources. In general, the inputs to the layout decision are as follows:

- ☞ Specification of the objectives and corresponding criteria to be used to evaluate the design. The amount of space required, and the distance that must be traveled between elements in the layout, are common basic criteria.
- ☞ Estimates of product or service demand on the system.
- ☞ Processing requirements in terms of number of operations and amount of flow between the elements in the layout.
- ☞ Space requirements for the elements in the layout.
- ☞ Space availability within the facility itself, or if this is a new facility, possible building configurations.

Objectives of Facility layout

The overall objective in designing a layout is to provide a smooth work flow and control; reducing cost of material through the factory or uncomplicated pattern for both consumers and workers in a service organization. Specific objectives of layout decision in service and manufacturing operations are outlined in the following section.

1. For manufacturing firm

- Provide enough production capacity
- Minimize material handling cost and effort
- Minimize labour requirements
- Provide a smooth flow of materials and product
- Maximize the use of available space

- Provide for volume and product flexibility and avoid bottleneck operations and contested areas
- Minimize health hazards
- Maximize the uses of machine tools.
- Provide communication opportunities for employees by positioning equipment and processes appropriately
- Maximize output
- Minimize supervisory and control requirements
- Ease of maintenance
- Provide space for personal – care needs and others

2. For service operations layout serves the following purposes

- provide for customer comfort and convenience
- allow attractive display
- reduce travel of personnel and customers
- provide for private in work areas
- promote communication
- provide for stock rotation for shelf life

Principles of Plant Layout

- 1. Principle of integration:** A good layout is one that integrates men, materials, machines and supporting services and others in order to get the optimum utilization of resources and maximum effectiveness.
- 2. Principle of minimum distance:** This principle is concerned with the minimum travel(or movement) of man and materials. The facilities should be arranged such that, the total distance travelled by the men and materials should be minimum and as far as possible straightline movement should be preferred.
- 3. Principle of cubic space utilization:** The good layout is one that utilizes both horizontal and vertical space. It is not only enough if only the floor space is utilized optimally but the third dimension, *i.e.*, the height is also to be utilized effectively.
- 4. Principle of flow:** A good layout is one that makes the materials to move in forward direction towards the completion stage, *i.e.*, there should not be any backtracking.
- 5. Principle of maximum flexibility:** The good layout is one that can be altered without much cost and time, *i.e.*, future requirements should be taken into account while designing the present layout.

- 6. Principle of safety, security and satisfaction:** A good layout is one that gives due consideration to workers safety and satisfaction and safeguards the plant and machinery against fire, theft, etc.
- 7. Principle of minimum handling:** A good layout is one that reduces the material handling to the minimum.

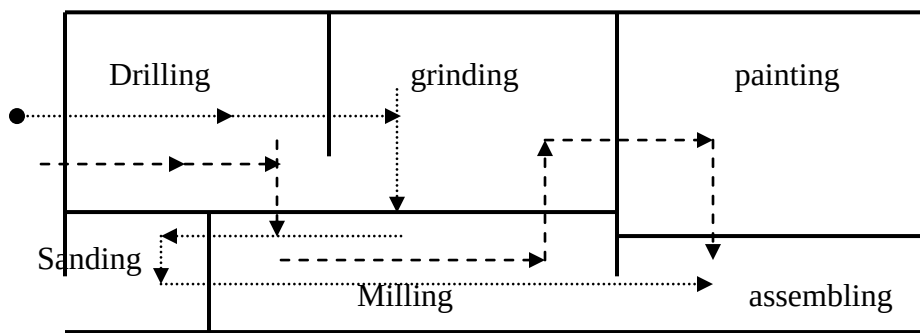
TYPES OF LAYOUT

The choice of layout type depends largely on process choices. Different layouts present managerial challenges, as well as different opportunities to satisfy unmet customer needs. There are three basic types (process layout, product layout, and fixed-position layout) and one hybrid type (group technology or cellular layout). In the following section we will see three basic type of layout along with their advantages and disadvantages.

i. Process layout

A process layout also called a *job-shop* or *functional layout*. In this Layout, machines are grouped according to similar functions into machine centers. Process layout is designed to process items or provide services that involve a variety of processing requirements. With a job process, which is best for low volume with high variety production, the operations manager must organize resources (employees and equipments) around the process. Process layout group departments/workstations according to functions or type of activities performed. Thus, all the resources that perform similar tasks are located together, so that materials can be routed through the resources in any order.

Generally, process layout consists of functional groupings of machines or labors that do similar works. For example, all drilling presses may be grouped together in one department and all milling machine in another (see the following figure). Depending on their processing requirements, parts may be moved to in different sequences among departments.



Advantages: Process layout provides the following benefits

- a. Lower initial capital investment in machines and equipments. There is high degree of machine utilization, as a machine is not blocked for a single product.
- b. The overhead costs are relatively low
- c. Systems are not vulnerable to equipment failure. That means breakdown of one machine does not result in complete work stoppage
- d. It is easy to avoid machine interference
- e. Because of its flexibility, process layout is less vulnerable to changes in product mix or design. Since it is flexible, idle equipment is usually available to replace machines that are temporarily out of service.
- f. Diversity of jobs offers more satisfaction to workers.
- g. It is possible to use individual incentive systems.

Disadvantages: Process layout suffers from following drawbacks

- a. Material handling costs are high due to backtracking and products may be moved frequently between departments.
- b. More skilled labour is required resulting in higher cost.
- c. Routing and scheduling is continual challenges because the level and type of work is highly variable
- d. Job complexity often reduces the span of supervision and results in higher supervisory costs than with product layout.
- e. Work in progress inventory is high needing greater storage space
- f. Processing rates tends to be slower (time gap or lag in production is higher)
- g. Productive time is lost in changing from one product or services to another.
- h. More frequent inspection is needed which results in costly supervision

Suitability: Process layout is adopted when

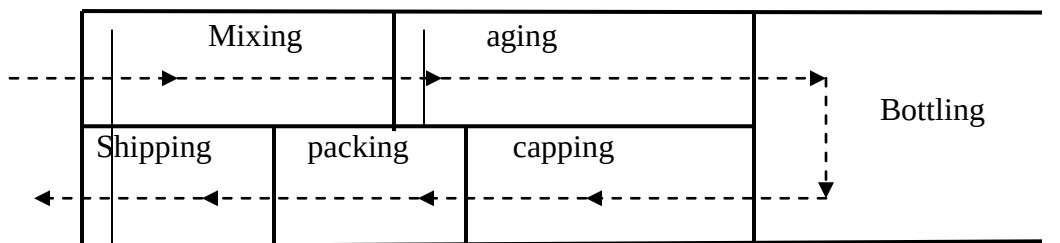
- a. Products are not standardized
- b. Quantity produced is small
- c. There are frequent changes in design and style of product
- d. Machines are very expensive

Thus, process layout or functional layout is suitable for job order production involving non-repetitive processes and customer specifications and non-standardized products, e.g. tailoring, light and heavy engineering products, made to order furniture industries, jewelry, etc

ii. **Product layout**

A product layout also called a flow-shop or line layout. Under this, machines and equipments are arranged in one line depending upon the sequence of operations required for the product. The materials move from one workstation to another sequentially without any back tracking or deviation. Therefore materials are fed into the first machine and finished goods travel automatically from machine to machine, the output of one machine becoming input of the next. The raw material moves very fast from one workstation to other stations with a minimum work in progress storage and material handling.

With line/continuous process, which are best for repetitive or continuous production, the operation managers dedicate resources to individual products or tasks. This strategy is achieved by product layout. In which work stations or departments are arranged in a linear path. In this case, resources are arranged around the products route rather than shared across many products. That is, equipments are arranged based on the sequence of operation, and products are move in a continuous path from one department to the next. It is common in high volume type of operations where products are standardized. Continuous flow (mass production) processing arrangements are usually organized by product layout. An example of product layout is winemaking which uses layout of this type as shown below.



Advantages: Product layout provides the following benefits:

- Faster processing rates due to mechanized fixed path material handling equipment and the machine pacing of the production rates.
- Low unit cost due to high volume: the high cost of specialized machine is spread over many units.
- Low cost of material handling, due to straight and short route and absence of backtracking
- Lesser investment in inventory and work in progress
- Smooth and uninterrupted operations

- f. Less congestion of work in the process
- g. Routing and scheduling are established in the initial design of the system; they do not require much attention once the system is operating.

Disadvantages: Product layout suffers from following drawbacks:

- a. High initial capital investment in special purpose machine
- b. Heavy overhead charges
- c. Breakdown of one machine will hamper the whole production process
- d. Difficult to avoid machine interference like excessive noise, vibration etc.
- e. Lesser flexibility as specially laid out for particular product.
- f. The jobs on production lines may provide little satisfaction to workers due to the high level of division of labor and the monotony that usually results.
- g. Difficulty of applying individual based incentive plans since the work is machine paced.

Suitability: Product layout is useful under following conditions:

- a. Mass production of standardized products
- b. Simple and repetitive manufacturing process
- c. Operation time for different process is more or less equal
- d. Reasonably stable demand for the product
- e. Continuous supply of materials

Therefore, the manufacturing units involving continuous manufacturing process, producing few standardized products continuously on the firm's own specifications and in anticipation of sales would prefer product layout e.g. chemicals, sugar, paper, rubber, refineries, cement, automobiles, food processing and electronics etc.

iii. Fixed position layout

In this type of layout, the major product being produced is fixed at one location. Equipment labour and components are moved to that location. All facilities are brought and arranged around one work center. The construction of large items, such as heavy machine tools, airplanes, buildings, power plants, dams and the like is usually accomplished in a fixed place. Rather than moving the item from one work centre to another, tools and components are brought to one place for assembly. Therefore, in a fixed positional

layout, the item being worked on remains stationary, and workers, materials, and equipments are moved about as needed. This is in marked contrast to product and process layouts. Almost always the nature of the product dictates this kind of arrangements: weight, size, bulk, or some other factor makes it undesirable or extremely difficult to move the product.

Advantages: Fixed position layout provides the following benefits

- a. It saves time and cost involved on the movement of work from one work station to another.
- b. The layout is flexible as change in job design and operation sequence can be easily incorporated.
- c. It is more economical when several orders in different stages of progress are being executed simultaneously.
- d. Adjustments can be made to meet shortage of materials or absence of workers by changing the sequence of operations.

Disadvantages: Fixed position layout has the following draw backs

- a. Production period being very long, capital investment is very heavy.
- b. As several operations are often carried out simultaneously, there is possibility of confusion and conflicts among different work groups.
- c. Very large space is required for storage of material and equipment near the product. For example, lack of storage space can present significant problems at a construction site in crowded urban locations.

Because of the many divers activity carried out on large projects and because of the wide range of skills required, special efforts are needed to coordinate the activities, and the span of control can be quite narrow. For this reasons the administrative burden is much higher than it would be under either of the other layout types. Fixed position layouts are widely used for farming, firefighting, road building, home building, remodeling and repair, and drilling for oil. In each case, compelling reasons bring workers, materials and equipments to the project's location instead of the other way around.

Suitability: The fixed position layout is followed in following conditions

- a. Manufacture of bulky and heavy products such as locomotives, ships, boilers, generators, wagon building, aircraft manufacturing, etc.
- b. Construction of building, flyovers, dams.
- c. Hospital, the medicines, doctors and nurses are taken to the patient (product).

iv. Group Technology (cellular) layout

There is a trend now to bring an element of flexibility into manufacturing system as regards to variation in batch sizes and sequence of operations. A *group technology (cellular) layout* groups dissimilar machines into work centers (or cells) to work on products that have similar shapes and processing requirements. Group technology (GT) layouts are now widely used in metal fabricating, computer chip manufacture, and assembly work. A group technology (GT) layout is similar to a process layout in that cells are designed to perform a specific set of processes, and it is similar to a product layout in that the cells are dedicated to a limited range of products. (*Group technology* also refers to the parts classification and coding system used to specify machine types that go into a cell.)

Group technology (GT) is the analysis and comparisons of items to group them into families with similar characteristics. GT can be used to develop a hybrid between pure process layout and pure flow line (product) layout. This technique is very useful for companies that produce variety of parts in small batches to enable them to take advantage and economics of flow line layout.

A group layout is possible where an item is being made in different types and sizes. Here machinery is arranged in a process layout but the process grouping is then arranged in a sequence to manufacture various types and sizes of products. It is to be noted that the sequence of operations remains same with the variety of products and sizes.

Thus **group layout** is a combination of the product layout and process layout. It combines the advantages of both layout systems. If there are m -machines and n -components, in a group layout, the m -machines and n -components will be divided into distinct number of machine-component cells (group) such that all the components assigned to a cell are almost processed within that cell itself. Here, the objective is to minimize the inter-cell movements.

The layout design process considers mostly a single objective while designing layouts. In process layout, the objective is to minimize the total cost of materials handling. Because of the nature of the layout, the cost of equipments will be the minimum in this type of layout. In product layout, the cost of materials handling will be at the absolute minimum. But the cost of equipments would not be at the minimum if the equipments are not fully utilized. In-group technology layout, the objective is to minimize the sum of the cost of transportation and the cost of equipments. So, this is called as multi-objective layout.

Advantages of Group Technology Layout

Group Technology layout can increase—

- ✱ Component standardization and rationalization.
- ✱ Reliability of estimates.
- ✱ Effective machine operation and productivity.
- ✱ Customer service.

It can decrease the—

- ✱ Paper work and overall production time.
- ✱ Work-in-progress and work movement.
- ✱ Overall cost.

Limitations of Group Technology Layout

This type of layout may not be feasible for all situations. If the product mix is completely dissimilar, then we may not have meaningful cell formation.

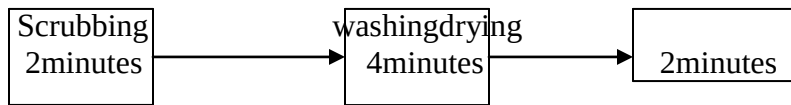
DESIGNING PRODUCT LAYOUT: LINE BALANCING

Line balancing is a procedure that can be used to optimize the assignment of tasks to work centers. The goal of line balancing is to obtain task groupings that represent approximately equal time requirements. This minimizes the idle time along the line and results in a high utilization of labor and equipment. Idle time occurs if task times are not equal among workstations; some stations are capable of producing at higher rate than others. The fast station will experience periodic waits for the output from slower stations or else be forced in to idleness to avoid buildups of work between stations.

Lines that are perfectly balanced will have a smooth flow of work as activities along the line are synchronized to achieve maximum utilization of labor and equipment. The major obstacle of attaining a perfectly balanced line is the difficulty of forming tasks bundles that have the same duration. There are different causes for this difficulty.

- a. It may not be feasible to combine certain activities in to the same bundle, either because of differences in equipment requirements or because the activities are not compatible.
- b. The differences among elemental tasks lengths cannot always be overcome by grouping tasks.
- c. An inability to perfectly balance a line is that a required technological sequence may prohibit otherwise desirable task combinations. Consider a series of three operations that have duration of two minutes, four minutes, and two minutes as shown in the following diagram. Ideally, the first and the third operations could be combined at one workstation and have a total time equal to that of the second

operation. However, it may not be possible to combine the first and the third operations. In the case of an automatic car wash, scrubbing and drying operations could not realistically, be combined at the same workstation due to the need to rinse cars between the two operations.



In real world, line balancing procedures are very complex and the procedures are heuristics. Line balancing heuristics do not guarantee optimal task assignments.

Terminologies in line balancing

- ✳ *Desired output*: the rate of output that is expected to be attained during operating time
- ✳ *Operating time*: total available time during specific period that used for operation
- ✳ *Cycle time*: is the maximum time allowed for work on a unit at each work station
- ✳ *Task*: is an element of work. E.g. grasping pen, positioning it on a paper, and writing.
- ✳ *Task length*: the amount of time required to complete a single task
- ✳ *Precedence relationship*: orders in which the tasks must be performed in the process
- ✳ *Assignment rule*: is rule by which individual tasks are going to be assigned to the work station
- ✳ *Work station*: a physical location where a particular set of tasks is performed
- ✳ *Work centre*: a physical location where two or more identical work stations are located
- ✳ *Productive time per hour*: the number of minutes in each hour that a work stations is working on the average.

Assignment rules

An assignment rule is a heuristic that establishes the bases for choosing an elemental task for assignment to a work station. There are five rules

- ✳ *longest task first*: assigning the task with the longest time first
- ✳ *shortest task first*: assigning the task with the shortest time first
- ✳ *most number following*: assigning the tasks with the largest number of followers first
- ✳ *least number following*: assigning the tasks with the least number subsequent tasks first
- ✳ *Ranked positional weight*: assign the tasks whose sum of task times of each following task is longest.

Steps in assembly line balancing

Step1. Draw the precedence diagram: The diagram consists of circles and arrows. Circle represents individual tasks and arrow indicates the order of task performance.

Step2. Find the required cycle time (C) using the formula:

$$C = \frac{\text{Operating time per day (OT)}}{\text{Desired output in unit (D)}}$$

Step3. Find the *theoretical minimum number of workstations (Nt)* using the formula:

$$Nt = \frac{\text{sum of task times (T)}}{\text{Cycle time (C)}} = \frac{\Sigma T}{C}$$

Step4. Select primary rule by which tasks are to be assigned to work station, and a Secondary rule to break ties.

Step5. Assign the task to work centers. The general rule is, assign tasks, one at a time, to the first work station until the sum of the task times is equal to the cycle time or no other tasks are feasible because of time or sequence restrictions. Repeat the process for work station 2, workstation 3 and so on until all tasks are assigned.

Step6. Evaluate the efficiency of the balance derived using the formula:

$$\text{Efficiency} = \frac{\text{sum of task times (T)}}{\text{Actual number of workstation (Na) X cycle time (C)}}$$

Balance delay (in percent) = 100 – efficiency

$$\text{Idle time} = (\text{number of workstation} \times \text{cycle time}) - \text{summation of task times}$$

Step7. If efficiency is unsatisfactory, rebalance using different decision rule.

Example:- The model J wagon is to be assembled on a conveyer belt. The assembly tasks that must be performed on each wagon are shown below. If 144 wagons are produced per hour, find the balance that minimizes the number of workstation, subject to cycle time and precedence constraints.

Assembly time and steps for model J wagon

Task time			Tasks that must
Task	(in seconds)	description	precede
A	12	position rear axle support and hand fasten 4 screws to nut	---
B	7	insert rear axle	A
C	8	tighten rear axle support screws to nuts	B
D	11	position front axle assembly and fasten with screw	---
E	8	tighten front axle assembly screw	D
F	12	position rear wheel #1 and fasten hubcap	E
G	5	position rear wheel#2 hubcap	C
H	10	position front wheel #1 and fasten hubcap	C

I 12 position front wheel #2 and fasten hubcap

F, G, H

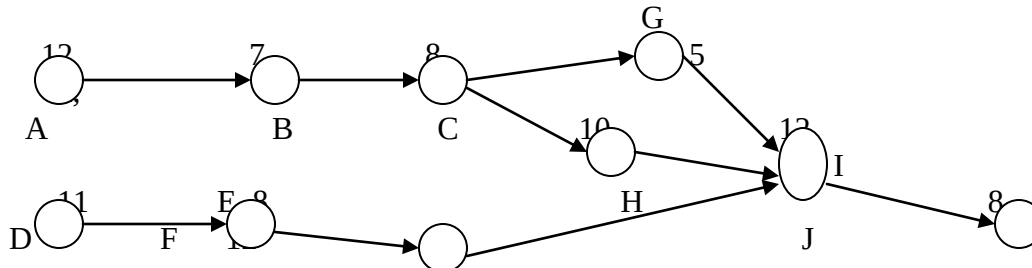
J 8 position wagon handle shaft on front axle assembly

and hand fasten bolt and nut

I

Solution

Step1. Draw the precedence diagram



Step 2 Determine cycle time: since time stated in seconds, first compute available production time in terms of seconds i.e. $60 \times 60 = 3600$ seconds. Then,

$$C = \frac{OT}{D} = \frac{3600}{144} = 25 \text{ seconds}$$

Step3. Find the theoretical minimum number of work station (Nt)

$$NT = \frac{\sum T}{C} = \frac{93}{25} = 3.72 \approx 4 \text{ workstation (rounded up)}$$

Step4. Select the assignment rule i.e. primary rule to assign the tasks and secondary rule to break ties if there is any.

Let us select largest positional weight as a primary rule and the most number of following as secondary rule.

Task	largest positional weight	number following
A	62	6
B	50	5
C	43	4
D	51	4
E	40	3
F	32	2
G	25	2
H	30	2
I	20	1
J	8	--

Step5. Make task assignments to form work station1, workstation 2 and so forth until all tasks are assigned.

Work station	available time	eligible (feasible) tasks	assigned	idle time
1	25	A,D	A	--
	13	B,D	D	--
	2	B,E	--	2
2	25	B,E	B	--
	18	C,E	C	--
	10	E,G,H	E	--
	2	F,G,H	--	2
3	25	F,G,H	F	--
	13	G,H	H	--
	3	G	--	3
4	25	G	G	--
	20	I	I	--
	8	J	J	--

Step6. Measure efficiency

$$\text{Efficiency} = \frac{\sum T}{N_a \times C} = \frac{93}{4 \times 25} = 93\%$$

$$\text{Balanced delay} = 100\% - 93\% = 7\% \text{ OR}$$

Balanced delay = 1- percentage of idle time

$$\text{Percentage of idle time} = \frac{\sum \text{idle time}}{N_a \times C} = 7/100 = .07$$

$$\text{Therefore balanced delay} = 1 - 0.07 = 0.93 = 93\%$$

What do we do if a task time is greater than cycle time?

For example, if the assembly line contains a task whose task times is 30 seconds, how do we deal with this task?

There are several ways that we may be able to accommodate the 30 second task in a 25-second cycle time.

- ✓ Split the task: can we split the task so that complete units are processed in two work station?
- ✓ Share the task: can the task somehow be shared so an adjacent work station does part of the work?
This differs from the split task in the first option because the adjacent station acts to assist, not to do some units containing the entire task.
- ✓ Use parallel work station it may be necessary to assign the task to two work stations that would appear in parallel.
- ✓ Use a more skilled worker: because this task exceeds the cycle time a faster worker may be able to meet the extra minutes required.
- ✓ Work over time and lastly, if possible redesign the product to reduce the task time slightly.

3.5 JOB DESIGN AND WORK MEASUREMENT

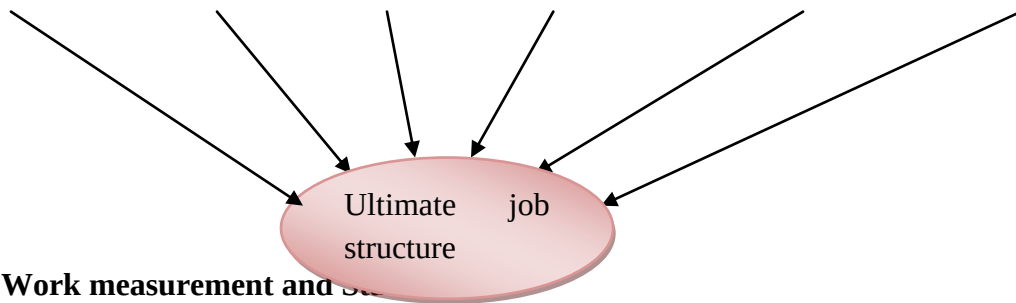
The operations manager's job, by definition, deals with managing the personnel that create a firm's products and services. To say that this is a challenging job in today's complex environment is an understatement. The diversity of the workforce's cultural and educational background, coupled with frequent organization restructuring, calls for a much higher level of people management skills than has been required in even the recent past. The objective in managing personnel is to obtain the highest productivity possible without sacrificing quality, service, or responsiveness. The operations manager uses job design techniques to structure the work so that it will meet both the physical and behavioral needs of the human worker. Work measurement methods are used to determine the most efficient means of performing a given task, as well as to set reasonable standards for performing it. People are motivated by many things, only one of which is financial reward. Operations managers can structure such rewards not only to motivate consistently high performance but also to reinforce the most important aspects of the job.

JOB DESIGN DECISIONS

Job design may be defined as the function of specifying the work activities of an individual or group in an organizational setting. Its objective is to develop job structures that meet the requirements of the organization and its technology and that satisfy the job holder's personal and individual requirements. The following figure summarizes the decisions involved.

Job design decisions

Who	What	Where	When	Why	How
Mental and the physical characteristics of the work force	Tasks to be performed	Geographic location of the organization; location of	Time of day; time of occurrence in the work flow	Organizational for the job; objectives and motivation of the worker	Methods of performance and motivation



Operations managers are interested in how long it takes to create an output or outcome, or equivalently, how much can be produced over a certain length of time. **Work measurement** is a *systematic procedure for the analysis of work and determination of times required to perform key tasks in processes*. Work measurement leads to the development of labor and equipment time standards that are used for

- estimating work-force and equipment capacity,
- establishing budgets,
- determining what new work procedures will cost,
- evaluating time and cost trade-offs among process design alternatives,
- establishing wage-incentive systems,
- monitoring and evaluating employee performance and productivity, and
- providing accurate information for scheduling and sequencing. Without accurate time standards it is impossible to perform these tasks.

The fundamental purpose of **work measurement** is to set time standards for a job. Such standards are necessary for four reasons:

1. **To schedule work and allocate capacity:** All scheduling approaches require some estimate of how much time it takes to do the work being scheduled.
2. **To provide an objective basis for motivating the workforce and measuring workers' performance:** Measured standards are particularly critical where output based incentive plans are employed.
3. **To bid for new contracts and to evaluate performance on existing ones:** Questions such as "Can we do it?" and "How are we doing?" presume the existence of standards.

4. **To provide benchmarks for improvement:** In addition to internal evaluation, benchmarking teams regularly compare work standards in their company with those of similar jobs in other organizations.

Work measurement is the process of creating labor standards based on the judgment of skilled observers. Managers often use informal methods to arrive at labor standards. They can develop simple estimates of the time required for activities or the number of employees needed for a job on the basis of experience and judgment. Properly set labor standards represent the amount of time that it should take an average employee to perform specific job activities under normal conditions. Labor/work standards are set in the following ways.

1. **Historical experience:** work standards can be estimated based on historical experience, that is, how many labor hours were required to do a task last time it was performed. Historical standards have the advantage of being relatively easy and inexpensive to obtain. They are usually available from employee time cards or production records. However, they are not objective, we do not know their accuracy, whether they represent a reasonable or a poor work pace, and whether unusual occurrences are included.
2. **Time studies:** a time study procedure involves timing a sample of worker's performance and using it as a basis for setting a standard time.

The general approach to time study can be described as follows.

Step 1: Selecting Work Elements (define the task to be studied): Each work element should have definite starting and stopping points to facilitate taking stopwatch readings. **Work elements that take less than three seconds to complete should be avoided because they are difficult to time.** The work elements selected should correspond to a standard work method that has been running smoothly for a period of time in a standard work environment. Incidental operations not normally involved in the task should be identified and separated from the repetitive work.

Step 2: Timing the Elements: After the work elements have been identified, the analyst times a worker trained in the work method to get an initial set of observations. The analyst may use either the *continuous method*, recording the stopwatch reading for each work element upon its completion, or the *snap-back method*, resetting the stopwatch to zero upon completion of each work element. For the latter method, the analyst uses two watches, one for recording the previous work element and the other for timing the current work element.

If the sample data include a single, isolated time that differs greatly from other times recorded for the same element, the analyst should investigate the cause of the variation. Time for an “irregular occurrence,” such as a dropped tool or a machine failure, should not be included in calculating the average time for the work element. The average observed time based only on representative times is called the **select time ($\bar{t}$)**. Irregular occurrences can be covered in the allowances.

CHAPTER FOUR

OPERATIONS PLANNING & CONTROL

4.1 Introduction

Planning is an integral part of a manager's job. If uncertainties cloud the planning horizon, it can be quite difficult for a manager to plan effectively. Firms plan their manufacturing and service operations activities at various levels and operate these as a system. Based on time dimension planning can be long range, medium range and short range.

4.1. Aggregate production planning

Aggregate plan: statement of a company's production rates, workforce levels, and inventory holding based estimates of customer requirements and capacity limitations.

Production plan: a manufacturing firm's aggregate plan, which generally focuses on production rate and inventory holdings. It determines the quantity and timing of production for the immediate future. *Aggregate planning* is an intermediate term planning decision. It is the process of planning the quantity and timing of output over the intermediate time horizon (3 months to one year). Within this range, the physical facilities are assumed to be fixed for the planning period. Therefore, fluctuations in demand must be met by varying labor and inventory schedule. Aggregate planning seeks the best combination to minimize costs.

APP is the process of planning the quantity and timing of production over an intermediate range by adjusting production rate, improvement and inventory. It is also translating annual and quarterly business plan into labor and production output plans for the intermediate term. The objective is to minimize the cost of resources required to meet demand over that period. The aim of aggregate planning is to get over all output levels in the near to medium future in the face fluctuating or uncertain demand. As a result of aggregate planning decisions and policies should be made concerning over time, hiring, layoff, subcontracting, and inventory levels. Aggregate planning determines not only the output levels planned but also appropriate resource input to be used. Aggregation on the supply side is by product families and on the demand side by groups of customers.

Main purpose of aggregate planning is to Specify the optimal combination of production rate, work force level, and inventory on hand. Aggregate planning is necessary in production and operations management because:

1. It facilities fully loaded facilities and minimizes overloading and under loading, thus keeping production cost low.
2. It provides adequate production capacity to meet expected aggregate demand
3. It facilitates the orderly and systematic transition of production capacity to meet the peaks and valleys of expected customer demand and;
4. In times of scarce production resources, it enhances the probability of getting the most output for the amount of resources available.

Steps in aggregate Planning

1. Begin with a sales forecast for each product that indicates the quantities to be sold in each time period (usually weeks, months or quarters) over the planning horizon (6 to 18months)
2. Total all of the individual product or service forecasts into one aggregate demand for a factory. If the products are not additive because of heterogeneous unit, a homogeneous unit of measurement must be selected that both allows the forecasts to be added and links aggregate outputs to production capacity.
3. Transform the aggregate demand for each time period into works, materials, machines, and other elements of production capacity required to satisfy aggregate demand.
4. Develop alternative resource schemes for supplying the necessary production capacity to support the cumulative aggregate demand.
5. Select the capacity plan from among the alternatives considered that satisfies aggregate demand and best meets the objectives of the organization.

Planning Strategies

There are four alternative strategies that deal with the workforce, work time, inventory and backlogs.

1. Vary the work force size by hiring and lying off employees as demand fluctuates.
2. Maintain a stable workforce, but vary the output rate by varying the number of hours worked through variable work weeks or overtime.
3. Maintain a stable workforce and constant output rate, but absorb demand fluctuations by allowing inventory to vary.
4. Allow backlogs (delivery lead time) to increase during periods of increased demand and decrease during periods of decreased demand.

The strategies can be applied independently or used in combination. This is known as mixed strategy which is common than individual strategies.

Proper Strategy is selected based on:

- 1) how much of each production resource is available
- 2) How much capacity is provided by each type of resource? The amount of resource required to produce a single unit of a particular product or service allows the translation of demand into production capacity plans.
- 3) At what step in production we determine capacity /labour hour available, or machine hour available
- 4) How much does it cost to scale capacity up or down cost of hiring, laying off, recalling/

AGGREGATE PRODUCTION PLANNING STRATEGIES

Managers often combine reactive (workforce adjustment, anticipation of inventory, workforce utilization, vacation schedules, subcontractors and backlogs, backorder and stocks) and aggressive (complementary products, and creative pricing) alternatives in various ways to arrive at an acceptable aggregate plan.

Chase strategy: a strategy that matches demand during the planning horizon by varying either the workforce level or the output rate. When a chase strategy uses the first method, *varying the workforce* level to match demand, it relies on just one reactive alternative-workforce variation. This chase strategy has the *advantage* of no inventory investment, overtime, or under-time. However it has some *drawback*, including the expense of continually adjusting workforce levels, the potential alienation of the workforce, and the loss of productivity and quality because of constant change in the workforce.

The second chase strategy, *varying the output rate* to match demand, opens up additional reactive alternatives beyond changing the workforce level. Sometimes called the *utilization strategy*, the extent and timing of the workforce's utilization is changed through overtime, under- time and vacation are taken. Subcontracting, including temporary help during the peak season, is another way of matching demand.

Level strategy: a strategy that maintains a constant workforce level or constant output rate during the planning horizon. When a level strategy used the first method, *maintaining a constant workforce level*, it might consist of not hiring or laying off workers (except at the beginning of the planning horizon), building up anticipation inventories to absorb seasonal demand fluctuations, using under-time in slack periods and overtime up to contracted limits for peak periods.

When level strategy uses the second method, *maintaining a constant output rate*, it allows hiring and layoffs in addition to other alternatives of first level strategy. The key to identifying a level strategy is whether the workforce or output rate is constant.

Mixed strategy: Strategies that consider and implements a full range of receive alternatives and goes beyond a “Pure” chase or level strategy. Whether management chooses a pure strategy or some mix the strategy should reflect the organizations environment and planning objectives.

Proper Strategy is selected based on:

- 1) how much of each production resource is available
- 2) How much capacity is provided by each type of resource? The amount of resource required to produce a single unit of a particular product or service allows the translation of demand into production capacity plans.
- 3) At what step in production we determine capacity /labour hour available, or machine hour available
- 4) How much does it cost to scale capacity up or down cost of hiring, laying off, recalling/

The Aggregate Production Process

Determining demand requirements: the first step in planning process is to determine the demand requirements for each period of planning horizon. The planner can derive future requirements for finished goods from backlogs (for make to order operations) or from forecasts for product families made to stock (for make to stock operations).

Identifying alternatives, constraints, and costs: the second step is to identify the alternatives, constraints, and costs for the plan. Constraints represent the physical limitations or managerial policies associated with the aggregate plan. Typically, many plans can satisfy specific set of constraints. The planner usually considers several types of costs when preparing aggregate plans:

- ⌚ Regular time costs (wages health insurance, dental care, social security, and retirement funds pay for vacation, holidays, and certain other types of absence).
- ⌚ Overtime costs:
- ⌚ Hiring and layoff costs
- ⌚ Inventory holding costs
- ⌚ Backorder and stock out costs

Preparing an acceptable plan: developing an acceptable plan is an iterative process i.e. plan may need to go through several revisions and adjustments. A prospective, or tentative, plan is developed to start. The plan must then be checked against constraints and evaluated in terms of strategic objectives.

Implementing and updating the plan: the final step is implementing and updating the final plan. Implementation requires the comment of manager in all functional areas. The planning committee may recommend changes in the plan during implementation or updating to balance conflicting objectives better. Acceptance of the plan does not necessarily mean that everyone is in total agreement, but it does imply that everyone will work to achieve it.

MATERIALS REQUIREMENTS PLANNING (MRP)

MRP is a means for determining the number of parts, components, and materials needed to produce a product. It provides time scheduling information specifying when each of the materials, parts and components should be ordered or produced. The basic purposes of MRP are to control inventory levels, assign operating priorities for items, and plan capacity to load the production system. The theme of MRP is *“getting the right materials to the right place at the right time.”*

BASIC MRP CONCEPTS

Independent demand: It exists when a demand for a particular item is unrelated to a demand for other item or when it is not a function of demand of other inventory item. Independent demands are not derivable or calculable from the demand of something else hence they must be forecast. For example, a work station may produce many parts that are unrelated but meet some external demand requirement.

Dependent demand: It is defined as dependent if the demand of an item is directly related to, or derived from the demand of another item or product. In dependant demand, the need for any one item is a direct result of the need for some other item, usually a higher level item of which it is part. In concept, dependent demand is relatively straight forward computational problem. Needed quantities dependant demand items are simply computed based on the number needed each higher level item in which it is used. MRP is the appropriate technique for determining quantities of dependent demand item. For example, if an automobile company plans to produce 50 cars per day, then obviously it will need 200 wheels and tires (plus spares). The number of wheels and tires needed is dependent on the production level and is not derived separately. The demand for car is independent.

Inputs to MRP

MRP is a processor which processes inputs (relating data) to give a time phased detailed schedule for raw materials and components. An MRP system has three major inputs:-

A. Master Production Schedule (MPS)

One of the three principal inputs of MRP system, the master production schedule, is a list of what end products are to be produced, how many of each product is to be produced, and when the products are to be ready for shipment. It is a driving input which an MRP system depends for its real effectiveness and usefulness because it is the determinant of future load, inventory investment, production, and delivery service. The MPS is derived from the aggregate schedule.

Lead time: In purchasing systems, lead time is the time between recognition of need for an order and receiving it. In production systems, it is the order, wait, move, queue, setup and runtimes for each component produced.

Cumulative lead time: The sum of the lead times that sequential phases of a process require from ordering of parts or raw materials to completion of final assembly.

B. *Bill of materials (BOM)*

Computation of the raw material and component requirements for end products listed in the master schedule is done by the product structure. The product structure is specified by the bill of materials, which is a structured list of all the component parts, assemblies and subassemblies that make up each product. A file which lists all assemblies together is the bill-of-materials file. Product structure tree is a visual depiction of the requirement in a BOM, where all components are listed by levels. Also a bill of material file identifies the specific materials used to make each item and the correct quantities of each.

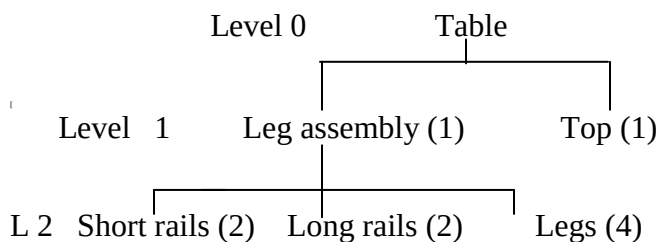


Figure: product structure for product table

For example leg assembly contains two of short nails and two of long nails. The other information contained in the bill-of-material file are the part number, child parent numbers, the date each child is to become effective or to be removed from use in the bill (effective date control for schedule engineering changes), dropout and yield percentages, shop floor delivery destination, and engineering revision level.

C. *Inventory records File (IRF) /Item Master File/*

One of the three primary inputs in MRP is inventory record file. It includes information on the status of each item by time period. It contains data such as the number of units on hand and on order. It comprises the individual item inventory records containing the status data required for the determination of net

requirements. This file is kept update by the position of inventory transactions which reflect the various inventory events taking place. Each transaction (stock receipt, disbursement, scrap, etc) changes the status of the respective inventory item. In addition to the status data, the inventory records also contain so-called planning factors used principally for the size and timing of planned orders.

Material Requirement Planning Process

Gross requirement (GR): it is the total amount required for a particular item. These requirements can be from external customer demand and also from demand calculated due to manufacturing requirements. GR is the projected needs for raw materials, components, subassemblies, or finished goods by the end of the period shown. Gross requirement comes from the master schedule (for end items) or from the combined needs of other items. But in MRP it is the quantity of item that will have to be disbursed, i.e. issued to support a parent order (or orders), rather than the total quantity of the end product.

Scheduled receipts (SR): represents the orders that have already been released and that are scheduled to arrive as of the beginning of the period. Once the paper work on an order has been released, what was prior to that event a planned order now becomes a scheduled receipt.

Projected available balance (PAB): it is the amount of inventory that is expected as of the beginning of a period.

This can be calculated as follows:

$$PAB_t = PAB_{t-1} - GR_{t-1} + SR_{t-1} + POR_{t-1} - \text{safety Stock}$$

On hand or available: the expected amount of inventory that will be on hand at the beginning of each time period.

Net requirements: is the amount needed when the projected available balance plus the schedule receipts in a period are not sufficient to cover the gross requirements.

Planned order receipt (POR) is the amount of an order that is required to meet a net requirement in the period. It is the quantity expected to be received by the beginning of the period in which it is shown under lot-for-lot (lot4lot) ordering, this quantity will equal net requirements. That is, under lot4lot ordering one can ask his supplier the exact quantity of the item needed for a particular time. Hence, the order size may vary depending on the requirement. Under lot-sizing ordering, this quantity may exceed net requirements. Any excess is added to available inventory in the next time period for simplicity, although in reality, it would be available in that period. Under lot-sizing ordering a buyer cannot vary the order size depending

on the requirements. Thus, each time equal quantity of item is ordered.

Planned order release (POR) is the planned order receipt offset by the lead time. This amount generates GR at the next level in the assembly or production chain.

Generally, the MRP processing takes the end item requirements specified by the MPS and “explodes” them into time phased requirements for parts, assemblies, components, and raw materials using the BOM and IRF offset by lead time.

EXERCISE

One unit of A is made of two units of B, three units of C and two units of D. B is composed of one units of E and two units of F. C is made of two units of F and one units of D. E is made of two units of D. Items A, C, D and F have one week lead time; B and E have lead times of 2 weeks. Item C has an on-hand (beginning) inventory of 15; D has on-hand inventory of 50; all other items have zero beginning inventory. We have schedule to receive 20 units of item E in week two; there are no other scheduled receipts.

Required

(A) Construct BOM (product structure tree)

(B) Prepare MRP table for each item; if 20 units of A are required in week 8.

4.2 OPERATIONS SCHEDULING

Scheduling is the processes of determining the starting and completion times to jobs. It is the determination of when labor, equipment and facilities are needed to produce a product or provide a service. Scheduling is a time table for performing activities, using resources, or allocating facilities. Generally, a schedule specifies the timing and sequence of production.

Schedule must be realistic; that is it must be capable of being achieved within the capacity limitations of the manufacturing facilities. Scheduling should be clearly differentiated from aggregate planning. The purpose of scheduling is to ensure that available capacity is efficiently and effectively used to achieve the organization’s objectives. The purpose of aggregate planning is to determine the resources (labor, equipment, space etc.) that should be acquired for scheduling. Often several jobs might be processed at one or more work stations. Typically a variety of tasks can be performed at each work station which make

effective scheduling a must rather than an alternative. Sequencing & loading should be considered during scheduling activity.

Sequencing: sequencing is concerned with developing an exact order (or sequence) of job processing. It is the determination of the order in which jobs are processed. One of the oldest sequencing methods is the Gant chart. Gant chart is a bar chart that shows a job's progress graphically or compares actual against planed performance.

Loading: is the assignment of work to specific resources/ machines. It is simply the process of assigning work to individual workers or machine. Loading can be: finite loading, infinite loading, back ward loading and forewarned loading.

Scheduling system can be either finite or infinite loading.

Loading refers to the assignment of jobs to processing (work) centers.

- ⌚ **Finite loading:** refers to loading activities with regard to capacity. Tasks are never loaded beyond capacity. Moreover, a finite loading approach actually schedules in detail each resource using the setup and runtime required for each order. In this case, the system determines exactly what will be done by each resource at every moment during the working day. If an operation is delayed due to a part or parts shortage, the order will sit in queue and wait until the part is available from a preceding operation.
- ⌚ **Infinite loading:** loading activity without regard to capacity. Moreover, infinite loading occurs when work is assigned to a work center simply based on what is needed over time. No consideration given directly to whether sufficient capacity at the resources required to complete the work, nor is actual sequence of the work as done by each resource in the work center considered.

Scheduling systems can also be generated forward or backward scheduling in time.

Back ward loading begins with the due date for each job and loads the processing time requirements against each work centre by proceeding back ward in time. The purpose of back ward loading is to calculate the capacity required in each work centre for each time period.

Foreward loading begins with the present date and loads jobs forwards in time. The processing time is accumulated against each work centre, assuming infinite capacity. The purpose of forewarned loading is to determine the approximate completion date of each job and the capacity required in each time period.

Objectives of scheduling

- ⌚ To meet the due date
- ⌚ To minimize job lateness
- ⌚ To minimize the lead time or setup time
- ⌚ To minimize work-in-process inventory
- ⌚ To maximize resource utilization
- ⌚ To provide the best customer service possible
- ⌚ Making the most efficient use of the people, equipments and facilities available to the organization.

Scheduling in manufacturing

Scheduling in manufacturing is the process of assigning priorities to manufacturing orders and allocating workloads to a specific work centers. Scheduling is challenging if the task variety is high. This is a case particularly for the job shop scheduling. In the following discussion, we will concentrate on scheduling issues for job shop production.

Job shop scheduling

For job shop production scheduling decision can be quite complex. What makes scheduling so difficult in a job shop is the variety of jobs (customer orders) that are processed, each with distinctive routing & processing requirement. In a pure job shop, there are several jobs to be processed, each of which may have different routing among department or machines in the shop. In designing a scheduling and control system for a high variety of activities, sequencing and prioritizing should be emphasized.

Priority rules for allocating jobs to machine- sequencing

As discussed above, sequencing is prioritizing jobs that have been assigned to limited resources. Sequencing is simple if work centers are lightly loaded and need the same processing time. But if work centers are heavily loaded there will be longer waiting time and idle time. In this case to minimize the waiting and idle time, we must prioritize the tasks by using priority rules.

Priority rules are the criteria by which the sequence of jobs is determined. The process of determine which job is started first on a particular machine or work centre is known as priority sequencing rules. Some of the most common priority rules for sequencing jobs are:

- ⌚ **First come, first served (FCFS):** orders are run in the order that they arrive in the department i.e. the oldest first rule.
- ⌚ **Shortest processing time (SPT):** run the job with the shortest completion time first i.e. shortest operating time first.
- ⌚ **Earliest due date first (due date):** run the job with earliest due date first. Thus, a job that is due tomorrow has a higher priority than the job that is due next week or next month.
- ⌚ **Critical ratio (CR):** this is calculated as the difference between the due date and the current date divided by the work remaining. Orders with the smallest CR are run first. In the CR rule, jobs are sequenced from lowest CR to highest CR. Those with a CR less than one are considered behind schedule and need to be expedited. And CR greater than one implies that the job is ahead schedule and can be de-expedite.

$$CR = \frac{\text{due date} - \text{today's date}}{\text{Total shop time remaining}} = \frac{\text{time remaining}}{\text{lead time remaining}}$$

- ⌚ **Last come first served (LCFS):** this rule occurs frequently by default. As orders arrive they are placed on the top of the stack and the operator usually picks up the order on top to run first.
- ⌚ **Slack time remaining (STR):** This is calculated as the difference between the times remaining before the due date minus the processing time remaining. Obviously, job with negative slacks are behind schedule; those jobs might require expediting to get caught up. Under the minimum slack rule, jobs are sequenced based on their slack: those with the most slack receive the lowest priority and those with the least slack receive the highest priority i.e. Orders with the shortest STR are run first.

Slack time remaining per operation (STR/OP):

$$STR/OP = \frac{\text{time remaining before due date} - \text{remaining processing time}}{\text{Number of remaining operations}}$$

Performance measures of job shop scheduling

From the operations manager's perspectives, identifying performance measures to be used in selecting a schedule is important. If the overall goals of the organizations are to be achieved the schedule should reflect managerially acceptable performance measure. The following describes the most common performance measure used in operation schedule.

- ⌚ *Job flow time*: refers to time a job spends in the shop. It is the sum of the moving time between operations, waiting time for machines or work orders, process time (including set up), and delays resulting from machine breakdowns, component unavailability and the like. The objective here is to minimize the average flow time.

$$\text{Flow time} = \text{waiting time} + \text{processing time}$$

$$\text{Average flow time} = (\text{sum of total flow time}) \div (\text{no of job processed})$$

- ⌚ *Makespan time*: refers to time to process a set of jobs. It is the total amount of time required to complete a group of job. It is the length of time between the start of the first job in the group and the completion of the last job in the group. The general objective is to minimize the make span time. Generally, makespan time = Σ processing time of each job
- ⌚ *Tardiness (past due)*: refers to the amount by which completion time exceeds the due date of a job. If a job is completed before its due date, tardiness is zero. The objective is to minimize the number of tardy jobs.

$$\text{Tardiness} = \Sigma \text{ past due, \& (Average tardiness} = \Sigma \text{times past due} \div \text{total no. of jobs})$$

- ⌚ *Work in process inventory*: any job in waiting line, moving from one operation to the next, being delayed for some reason, being processed and the like are work in process. This measure can be expressed in units, number of jobs, and birr value for the entire system and so on.

$$(\text{Average WIP inventory} = \Sigma \text{job flow times} \div \text{makespan times})$$

- ⌚ *Total inventory in the system*: is the sum of the schedule receipt and on hand inventories. Average total inventory = Σ jobs in the system $\div$ makespan time
- ⌚ *Utilization* = total processing time $\div$ total flow time
- ⌚ *Average number of jobs in the system* = $\frac{\Sigma(\text{no. of jobs}) (\text{production time of the job})}{\text{Makes pan time}}$

Makes pan time

Job shop scheduling techniques and priority rules

There are a number of job scheduling techniques. However, we will take two most widely practiced job scheduling techniques: *scheduling 'n' jobs on one machine problems and scheduling 'n' jobs on two machines.*

i. scheduling n jobs on one machine(n/1cases).

Let us compare some of the priority rules in a static scheduling situation involving some jobs and one machine. The theoretical difficulty of this type of problem increases as more machines are considered.

Example 1

Mesfin industrial engineering has engaged in assembling different model cars. Assume five customers submitted their orders for different model of cars: the first customer order model type A, the second model type B, the third model type C, the fourth for model type D and the fifth order model type E. Further assume that all order requires the use of only one machine which forces the company to decide on the processing sequence for the five orders. The following is the time (in days) required to complete the job on in the machine.

<u>Job (in order of arrival)</u>	<u>processing time (in days)</u>	<u>No .of days till due or due date (days hence)</u>
Model A	3	5
Model B	4	6
Model C	2	7
Model D	6	9
Model E	1	2

Required: based on FCFS rule, determine makespan time, total flow time, average flow time, average tardiness (average job lateness), average WIP inventory, average total inventory, utilization & no. of jobs in the system.

Solution

The FCFS rule results in the following computation.

Job order	Begin work	processing Time (in days)	job flow time	scheduled customers pick up time	actual customer pick up time	days early	days past due
A	0	3	3	5	5	2	-
B	3	4	7	6	7	-	1
C	7	2	9	7	9	-	2
D	9	6	15	9	15	-	6
E	15	1	16	2	16	-	14

Make span time = $3+4+2+6+1= 16$

Total flow time= $3+7+9+15+16=50$

Average total flow time= $50 \div 5 = 10$ days

Average tardiness= $0+1+2+6+14 \div 5 = 4.6$ days

Average WIP inventory= $50 \div 16 = 3.125$

Average total inventory = $(5+7+9+15+16) \div 16= 3.25$

Utilization = $16 \div 50 = 0.32= 32\%$

$$\text{No. of jobs in the system} = (3 \times 5) + (4 \times 4) + (2 \times 3) + (6 \times 2) + (1 \times 1) \div 16 = 3.125$$

Example 2: ABC factory is engaged in gentle men garment business. The order receiving unit has received the orders from its customers for coats, gowns, shirts, trousers, and jackets respectively. There is only one job centre that all kinds of jobs are performed. The following is the time (in days) required to complete the job in the work centre.

Job order	Processing time(in days)	No of days till due
Coats(C)	7	8
Gown (G)	2	3
Shirt(S)	5	7
Trouser (T)	3	9
Jacket(J)	6	6

Required:

- Using CR priority rule, schedule the jobs.
- Evaluate the in terms of performance measure i.e. makespan time, average day early, average job lateness, average WIP inventory, average total inventory utilization and no. of jobs in the system.

Order	No of days till due (a)	processing time (b)	CR= (a÷ b)
C	8	7	1.14
G	3	2	1.5
S	7	5	1.4
T	9	3	3
J	6	6	1

From the highest critical to lowest critical ratio, the job can be scheduled as: J, C, S, G, T in the order of 1st, 2nd, 3rd, 4th, 5th respectively.

Job order	Begin work	processing Time(in days)	job flow time	scheduled customers pick up time	actual customer pick up time	days early	days past due
J	0	6	6	6	6	-	-
C	6	7	13	8	13	-	5
S	13	5	18	7	18	-	11
G	18	2	20	3	20	-	17
T	20	3	23	9	23	-	14

Now , you can determine make span time, total flow time, average flow time, average tardiness (average job lateness), average WIP inventory, average total inventory, utilization & no. of jobs in the system as follows.

Make span time = 6+7+5+2+3=23 days

Total flow time = 6+13+ 18+ 20+23= 80 days

Average flow time = $80 \div 5 = 16$ days

Average tardiness = $(5 + 11 + 17 + 14) \div 5 = 9.4$ days

Average WIP inventory = $80 \div 23 = 3.478$

Average inventory = $(6 + 13 + 18 + 20 + 23) \div 23 = 3.478$

Utilization = $23 \div 80 = 0.2875 = 28.75\%$

No. of jobs in the system = $(6 \times 5) + (7 \times 4) + (5 \times 3) + (2 \times 2) + (3 \times 1) \div 23 = 3.478$

Exercise: Based on the above data, determine make span time, total flow time, average flow time, average tardiness (average job lateness), average WIP inventory, average total inventory, utilization & no. of jobs in the system using:

i. FCFS ii. LCFS iii. EDD, iv. SPT v. Random order

Scheduling n jobs on two machine

The next set up in complexity of job shop type is the n/2 cases, where two or more jobs must be processed on two machines in common sequences. As in n/1 cases, there is an approach that leads to an optimal solutions according to certain criteria. Also as in n/1 cases, we assume that it is a statistic scheduling situation. The objective of this approach, termed Johnson's rule or method (after its developer), is to minimize the flow time, from the beginning of the first job until the completion of the last. Johnson's rule is an algorithm for sequencing any number of jobs through two serial operations to minimize makes span time and total flow time. Johnson's rule consists of the following steps:

Step1. List the operation time for each job on both machines.

Step2. Select the job with the shortest operation time.

Step 3. if the shortest time is for the first machine, do the job first; if the shortest time is for the second machine, do the job last.

Step4. Eliminate the job assigned under step 3

Step5. Repeat step 2 and 3 for each remaining job until the schedule is complete.

N: B if there is a tie, i.e. if a job has the same processing time at each work centre, it makes no differences whether we place it forward the beginning or the end of the sequence so we can assign arbitrarily.

Example1:

A wood carpentry has received orders from its customers for chair, table shelf, door, and board. The owner has organized two job centers one for assembly and one for finishing. Every product has to pass through the two work centers in the sequence. The following is the time (per hour) required to complete the jobs in each work centre.

Job order	assembly (in hours)	finishing (in hours)
Chair(C)	4	2
Table(T)	3	8
Shelf(S)	4	1
Door(D)	6	3
Board(B)	5	9

Required:

- schedule the work centre using FCFS and determine make span, total flow time, total idle time, and average flow time
- schedule the work centre using Johnson rule and then determine make span, total flow time, total idle time, and average flow time.

Solution

- using FCFS rules: the sequence is C - T - S - D & B

	0	4	7	11	17	22	31	
Ass.	C=4	T=3		S=4	D=6		B=5	Free for other task
Fin.	I=4	C=2	I=1	T=8		S=1	I=1	D=3
							I=2	B=9

Make span time= $4+3+4+6+5+9=31$ hrs

Total flow time= $4+7+11+17+22+31=92$ hrs

Total idle time = $4+1+1+2= 8$ hrs

b. scheduling using Johnson rule

First determine job sequence using Johnson's rule as discussed above.

1 st	2 nd	3 rd	4 th	5 th
T	B	D	C	S

	0	3	8	14	18	22	26	
Second job as refers to idle)	Ass.	T=3	B=5	D=6	C=4	S=4	Free other task	
	Fin.	I=3	T=8	B=9		D=3	C=2	S=1

schedule the follows (I

Make span time = $3+5+6+4+4+1+2+1=26$ hrs

TFT = $0+3+8+14+18+22+26=91$ hrs

Average flow time = $91 \div 5 = 18.2$ hrs

Total idle time = 3 hours

Exercise: the Morris machine company just received an order to refurbish five motors for materials handling equipment that were demanded in a fire. The motors will be repaired at two work stations in the following manner.

Work station 1 = dismantling the motor and clean parts

Work station 2 = replace parts as necessary, test the motors and make adjustments

The customer's shop will be inoperable until all the motors have been repaired the estimated time for repairing each motor is shown in the following table.

Motor	workstation 1	work station 2
M1	12	22
M2	4	5
M3	5	3
M4	15	16
M5	10	8

Develop the schedule that minimize the make span time using Johnson's rule.

1.1. Scheduling in services

Scheduling is also important in service organizations. For example nurses must be scheduled in hospital, and truck must be scheduled for deliveries for furniture distributors. One important distinction between manufacturing and services that affects scheduling is that service operations cannot create inventories to buffer demand uncertainties. A second distinction is that in service operations demand often is less predictable; customers may decide on the spur of the moment that they need a hamburger, a hair cut or a plumbing repair. Thus capacity, often in the form of employees is crucial for service providers. In this section we discuss various ways in which schedule systems can facilitate the capacity management of service providers.

Scheduling customer demand

One way to manage capacity is to schedule customers for arrival times and definite periods of service time. With this approach, capacity remains fixed and demand is leveled to provide timely service and utilize capacity. Three methods are commonly used: appointment, reservation and backlog.

Appointment: an appointment system assigns specific times for services to customers. The advantages of this method are timely customer service and high utilization of servers. Doctors, dentists, lawyers, and automobile repair shops are examples of service providers that use appointment systems. Doctors can use the system to schedule parts of their days to visits hospital patients and lawyers can set aside time to prepare cases. If timely service is to be provided, however, care must be taken to tailor the length of appointment to individual customer needs rather than merely scheduling customers at equal time interval.

Reservation: reservation system, although quite similar to appointment systems, are used when the customer actually occupies or uses facilities associated with the service. For example, customer reserve hotel rooms, automobiles, airline seats, and concert seats. The major advantage of reservation system is the lead time they give service managers to plan the efficient use of facilities. Often, reservation requires require some form of down payment to reduce the problem of no-shows

Backlog: a less precise way to schedule customers is to allow backlogs to develop; that is customers never know exactly when service will commence. They present service request to an order taker, who adds it to

the waiting line of orders already in the system. TV repair shops, restaurants, banks, grocery stores, and barber shops are examples of the many types of business that uses this system. Various priority rules can be used to determine which order to process next. The usual rule is first come, first served, but if the order involves rework on a previous order, it may get a higher priority.

Scheduling the worker:

Another way to manage capacity with a schedule system is to specify the on duty and off duty periods for each employee over a certain time period as in assigning postal clerks, nurses, pilots, attendants, or police officers to specific work days and shifts. This approach is used when customers demand quick response and total demand can be forecasted with reasonable accuracy. In this instant capacity is adjusted to meet the expected loads on the service system. The work force capacity available each day must meet or exceed daily work force requirements. If it does not, the scheduler must try to rearrange days off until the requirements are met. If no such scheduling can be found, management might have to change the staffing plan and authorize more employees, over time hours or large backlogs.

Managers usually use rotating schedule than fixed schedule to assign workers on their duty. In rotating schedule rotate employees through a series of work days or hours. Thus over a period of time, each person has the same opportunity to have weekends and holidays off and to work days, as well as evening and nights. A rotating schedule gives each employee the next employee's schedule the following week. In contrast, a fixed schedule calls for each employee to work the same days and hours each week.

CHAPTER FIVE

QUALITY MANAGEMENT AND CONTROL

5.1 MEANING AND NATURE OF QUALITY

Different meaning could be attached to the word quality under different circumstances. The word quality does not mean the quality of manufactured product only. It may refer to the quality of the process (*i.e.* men, material, and machines) and even that of management. Where the quality manufactured product referred as or defined as “Quality of product as the degree in which it fulfills the requirement of the customer. It is not absolute but it judged or realized by comparing it with some standards”.

Quality begins with the design of a product in accordance with the customer specification further it involved the established measurement standards, the use of proper material, selection of suitable manufacturing process etc., quality is a relative term and it is generally used with reference to the end use of the product.

Crosby defined as “Quality is conformance to requirement or specifications”.

Juran defined as “Quality is fitness for use”. “The Quality of a product or service is the fitness of that product or service for meeting or exceeding its intended use as required by the customer.”

While quality management is cross functional in nature and involves the entire organization, operations have special responsibility to produce a quality product for the customer. This requires the cooperation of the entire organization and careful attention to management and control of quality.

Quality can be defined in a number of different ways.

In an article entitled “what does product quality really mean?” David Garvin, discuss five approaches to define quality:

- ☞ He state that a common notion of quality is that it is synonymous with ‘*superiority*’ or ‘*innate excellence*’. From this view point, quality cannot be precisely defined but can only be recognized through experience.
- ☞ The second view point is that quality is precise and measurable concept and that *differences in quality* reflect differences in quantity of some *product attributes*. For instance, high quality ice cream has high butterfat content. According to this approach, quality can be ranked based on the amounts of the desired attributes they possess. (Product based view)
- ☞ The third view is that quality is determined by what a customer wants and is willing to pay. According to this view the goods and services that *best satisfy* individual consumer’s unique needs or wants are

regarded as having the highest quality. Thus, quality can be defined as *fitness for intended use*, or in other words, how well the product performs its intended functions. (user based view)

☞ The fourth definition of quality arises from the unique perspective of manufacturing operations. In this setting quality is associated with engineering and manufacturing practices; hence, the perspective of quality is synonymous with *conformance to specification*. Quality conformance can be defined as, how well manufacturing is able to meet design specification.(manufacturing view)

☞ Finally the value based approach to quality defines it in terms of *cost and price*. In this sense, a quality product is one that provides a predetermined level of performance at an acceptable price or provide conformance to design specification at an acceptable cost. (Value based approach)

In considering all these approaches to quality, it is clear that the meaning of quality depends on one's view point and places in the organization. Thus different definitions are needed. More over it is necessary to shift one's perspective of quality as products move from their design stage to market. All of the viewpoints just presented are necessary in order to result in an overall quality product.

Quality Assurance Vs Strategic Approach

✚ **Quality Assurance** is concerned with the entire range of production, beginning with product or service design, continuing through the transformation process and extended to service after delivery. It emphasis on finding and correcting defects before reaching market.

✚ **Strategic approach** is proactive, focusing on preventive mistakes from occurring. Greater emphasis on customer satisfaction

Dimensions of Quality

- **Performance:** main characteristics of the product or services.
- **Aesthetics:** appearance, feel, smell, taste
- **Special feature:** extra characteristics.
- **Conformance:** how well a product or service corresponds to the customer's expectation
- **Safety:** risk of injury or harm
- **Reliability:** consistency of performance.
- **Durability:** the useful life of the product or service.
- **Perceived quality:** indirect evaluation of quality (e.g. reputation)
- **Service after sale:** handling of complaints or checking on customer satisfaction.

Dimension of Quality: Service

- **Time and timeliness:** How long a customer waits for service, and is it completed on time?

- **Completeness:** is everything customer asked for provided?
- **Courtesy:** involves politeness, respect, consideration, and friendliness of contact personal (including receptionists, telephone operators, etc). How are customers treated by employees?
E.g. Are catalogue phone operators nice and are their voices pleasant?
- **Consistency:** Is the same level of service provided to each customer each time?
- **Accessibility and convenience:** -involves approachability and ease of contact.
 - How easy is it to obtain service?
 - Does a service representative answer your calls quickly?
- **Accuracy:** is the service performed right every time?
- **Responsiveness:** how well does the company reach to unusual situation?

5.2. OVERVIEW OF TQM

Total Quality Management is a philosophy that involves each and every individual in an organization in a continual effort to improve quality and achieve customer satisfaction. Total quality management (TQM) is an integrated organizational effort designed to improve quality at every level.

The TQM Approach

TQM is not called philosophy for nothing. It is that common viewpoint as well as attitude shared by the whole organization that helps the organization achieves its prime objective of increase in revenue as well as a continuous relationship with the customer, by providing a quality based service which fulfills the customer's needs and requirements. Avert

If we apply the TQM approach we can identify the role played by various departments and interfaces of the organization. These roles at the functional and departmental levels if not in line with the organizational strategy would not allow the organization to pursue TQM.

The Evolution of Total Quality Management (TQM)

The concept of quality has existed for many years, though its meaning has changed and evolved over time. In the early twentieth century, quality management meant inspecting products to ensure that they met specifications. In the 1940s, during World War II, quality became more statistical in nature. Statistical sampling techniques were used to evaluate quality, and quality control charts were used to monitor the production process. In the 1960s, with the help of so-called "quality gurus," the concept took on a broader meaning. Quality began to be viewed as something that encompassed the entire organization, not only the production process. Since all functions were responsible for product quality and all shared the costs of poor quality, quality was seen as a concept that affected the entire organization.

The meaning of quality for businesses changed dramatically in the late 1970s. Before then quality was still viewed as something that needed to be inspected and corrected. However, in the 1970s and 1980s many U.S. industries lost market share to foreign competition. In the auto industry, manufacturers such as Toyota and Honda became major players. In the consumer goods market, companies such as Toshiba and Sony led the way. These foreign competitors were producing lower-priced products with considerably higher quality.

To survive, companies had to make major changes in their quality programs. Many hired consultants and instituted quality training programs for their employees. A new concept of quality was emerging. One result is that quality began to have a strategic meaning. Today, successful companies understand that quality provides a competitive advantage. They put the customer first and define quality as meeting or exceeding customer expectations.

Since the 1970s, competition based on quality has grown in importance and has generated tremendous interest, concern, and enthusiasm. Companies in every line of business are focusing on improving quality in order to be more competitive. In many industries quality excellence has become a standard for doing business. Companies that do not meet this standard simply will not survive. As you will see later in the chapter, the importance of quality is demonstrated by national quality awards and quality certifications that are coveted by businesses.

The term used for today's new concept of quality is *total quality management* or *TQM*. Figure presents a timeline of the old and new concepts of quality. You can see that the old concept is *reactive*, designed to correct quality problems after they occur. The new concept is *proactive*, designed to build quality into the product and process design.

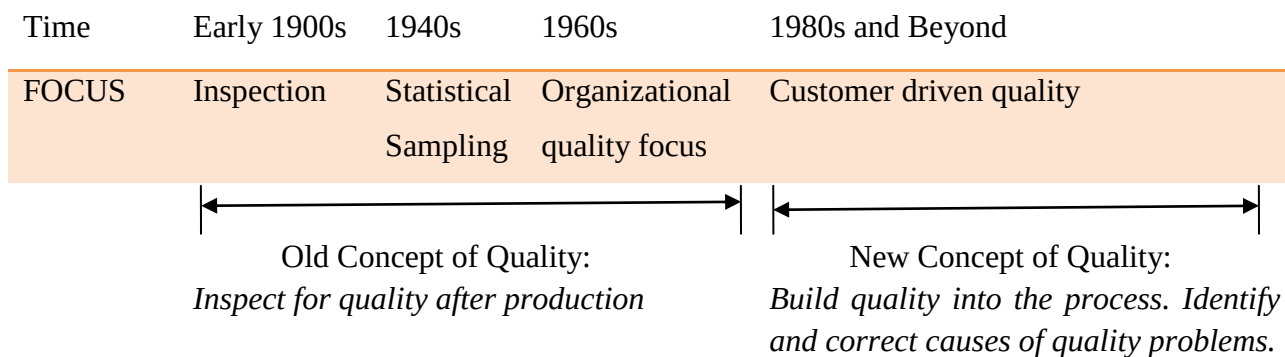


Figure 5-1 Timeline showing the differences between old and new concepts of quality

COST OF QUALITY

A new idea in the quality area is to calculate and control the cost of quality. The cost of quality may be divided into two components: *control cost* and *failure cost*. The control costs are related to activities which remove defects in the production stream. This can be done in two ways: by *prevention* and by *appraisal*.

A. Control cost

i. The prevention cost

Prevention costs are costs associated with preventing defects before they happen. Such costs include:

- ☞ Cost of providing quality engineering, and quality planning services for ensuring correct specifications of materials, use of right methods and process, preparing company standards, preparing sampling procedures and so on.
- ☞ Information Costs: Costs of acquiring and maintaining data related to quality and development of reports on quality performance.
- ☞ Cost involved with training and retraining of operators, supervisors and other staffs.
- ☞ Cost of research and development efforts, so as to maintain high quality products.
- ☞ The cost of redesigning the process to remove the cause of poor quality.
- ☞ Cost incurred in the organization of quality circles and other techniques with the objectives of creating interests and involvements of workers and staffs in their work motivate them and high quality of work life. These activities occur prior to production and are aimed at preventing defects before they occur.

ii. The appraisal (inspection)

Appraisal costs are costs incurred in assessing the level of quality attended by the operating system. Appraisal helps management identify quality problem. Such costs include:

- ✓ Cost of testing or inspecting incoming raw materials, including the cost of their movement for the purpose of inspection testing at regular interval
- ✓ Cost of providing and maintaining laboratory services for the purpose of inspection
- ✓ Cost of process control test or stage inspection.
- ✓ Cost of product inspection, such as mechanical testing, non destructive testing, cost of carrying out field trials etc.
- ✓ Cost of maintenance and calibration of test and inspection equipment and apparatus at regular interval.
- ✓ Expenditure incurred in vendor rating when any of the materials required for the product are procured from outside sources.

B. Failure Cost

The failure costs are incurred either during the production process (internal) or after the production is shipped (external).

i. The internal failure

Internal failure costs result from defects that are discovered during the production of a product or services. Such costs include:

- ✓ Cost of scrap or rejections produced which cannot be passed on to, or which will not be accepted by the customer and which becomes a total cost. It will include the cost of power and various in process materials spent in producing the rejection.
- ✓ Cost of rework or corrective operations, in case of such items which have not been passed during inspection but which can be made acceptable after certain rework or repair such as welding, brazing, pressing, filling, re-heat treatment, rough machining etc.
- ✓ Cost involved in fault investigation, trouble –shootings, defect analysis. It may also entail cost of re-examination, and testing, test methods, change of material specification or method of production etc.
- ✓ Loss in capacity of production because of the rejection produced.

ii. The external failure cost

External failure costs arise when a defect is discovered after the customer has received defective products or services. Such costs include product:

- ✓ Loss of future orders to the company owing to loss in its prestige caused by high rejection or poor performance in services. The customer may even withhold payments and the relation may be impaired which may be difficult to improve again. The customers may permanently withdraw placing order.
- ✓ Cost involved in attending to customers complaints and providing customer services, including warranty charges (the cost of refund, repair, or replace), returned merchandises (cost related to returning goods to sellers including transportation), losses of taxes and duties, allowance (cost of concession), complaints (the cost of setting customer complaints) and the like.
- ✓ Litigation costs which include not only legal fees but also the time and effort of employees who must appear for the company in court.

The total cost of quality can thus be expressed as the sum of the following cost:

$$\text{Total cost} = \text{control cost} + \text{failure cost, that is} \\ (\text{Prevention cost} + \text{appraisal cost}) + (\text{internal failure cost} + \text{external failure cost})$$

The total cost of quality can be minimized by observing the relationship between the cost of quality and degree of conformance. When the degree of conformance is very high (low defects) the cost of failure are low but the cost of control are quite high. When the degree of conformance is low (high defect) the opposite situation exist. Good quality management requires the proper balance between appraisal and prevention costs, so that total control costs are at the minimum.

5.3 Continuous Improvement

Because of advances in process capability, nonconformity prevention eventually loses its appeals as focus quality improvement efforts. The ability to satisfy customer's requirement today, however, does not imply an ability to satisfy customer's expectation in the future, because customer expectation change over time. Nor does the fact that, a process is capable of producing without non conformities means it cannot be improved. Finally, processes that are not improving may be eroding.

For all these reasons, the focus of quality improvement efforts may eventually shift from prevention to ongoing and continuous improvement. Focusing on ongoing improvement is a way of preparing to meet customer's future expectations, whether it can be anticipated or not. Even when **future customer's expectations** are **unpredicted and unforeseeable**, a focus on continual improvement can lead to the development of capabilities that will prepare a firm to meet new expectations. Ongoing, continuous improvement of processes requires a company to establish a climate in which everyone focuses on delighting the customers. Thus, the tools and methods that are used to foster ongoing improvement are meant not just for quality professional, but for everyone. They are most useful in the context of an organization wide focus on customer satisfaction consistent with four commitment of total quality management.

5.4 Statistical Quality Controls

One of the cornerstones of quality control is the use of statistical methods to determine how much inspection to use. **Statistical quality controls are the general category of statistical tools used to evaluate organizational quality.** In many case a great deal can be saved by **taking a sample** rather than making 100 percent inspection. In other cases there is no alternative but to take a sample. For example, destructive testing. Two distinct types of statistical methods are available: *acceptance sampling and process control*. Consider each in the following discussion.

i. Acceptance sampling

Applies to group inspection where a decision to accept or reject a lot of materials is made on the basis of **random sample** drawn from the lot. This type of inspection is frequently **used for incoming raw materials** or for **finished goods** prior to shipment.

Generally it can be defined as taking one or more samples at random from a lot of items, inspecting each of the items in the sample (s) and deciding – on the basis of inspection result- whether to accept or reject the entire lot. **This type of inspection can be used by the customers to ensure that quality standards are met prior to shipments.** Acceptance sampling is used in preference to 100% inspection where ever the cost of inspection is high in relation to the cost of passing defective items to the customer.

In a single acceptance sampling, one sample is taken from a lot and the decision whether to accept or reject the lot is made after the sample is inspected and compared with standards. Formally, we let: n = sample size, c = acceptance number, x = number of defective units found in the sample. For single sampling, the decision rule whether to accept or reject the lot after inspecting the sample is as follows:

If $x \leq c$, accept the lot

If $x > c$, reject the lot

For example, suppose we have a lot of 10,000 items and we decided to take a random sample of 100 items ($n=100$). We inspect the 100 items and find 3 defectives ($x=3$). Assume the acceptance number in this case is 2 ($c=2$). Since the number of defective units in the sample exceeds the acceptance number, the lot of 10,000 units will be rejected. Note that very good lots or very bad lots will usually require only one sample and lots of medium quality may require two or more samples to reach a decision.

ii. Process quality control system

No two products or services are exactly alike because the processes used to produce them contain many **sources of variation**, even if the processes are working as intended. For example, the time required to process a credit card application varies because of the load on the credit department, the financial background of the applicant, and the **skill & attitude of the employees**. Nothing can be done to eliminate variation in process output completely, but management can investigate the cause of variation. Generally, **the source of variation can be common or random causes of variation and assignable causes.**

a. Common causes of variation

Common causes of variation are **purely random, unidentifiable sources of variation that are unavoidable** with the current process. No matter how perfectly the process is designed, there will be some

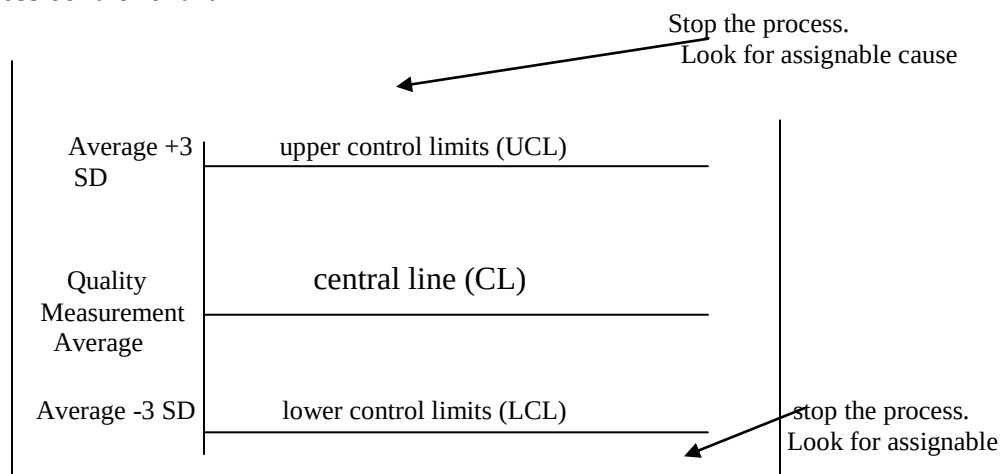
variability in quality characteristics from one unit to the next. For example, a machine filling cereal boxes will not deposit exactly the same weight in each boxes; the amount filled will vary around some average figure. **The aim of process control is to find the range of *natural variation* of the process and to then ensure that production stays within this range.** Natural variation is usually under the state of control.

b. Assignable causes

The second category of variation is assignable sources of variation also **called *special cases*** includes any abnormal variations which are **not usually found in a state of control**. Assignable causes of variation are any variation causing factor that can **be identified and eliminated**. Assignable causes that results abnormal variation may **include: lax (careless) procedures, untrained operators, improper machine maintenance**. The first job of process control manager is to **seek out these sources of unnecessary variation** and bring the process under statistical control, where the remaining variation is due to random causes.

A process can be brought to a state of control and can be maintained in this state through the use of quality control charts –also called *process chart or control chart*. In the control chart shown below the Y axis represents the quality characteristics which is being controlled while the X axis represent time or particular sample taken from the process. The center line of the chart is the average quality characteristic being measured. The upper control limit represents the maximum acceptable random variations and the lower control limit indicates the minimum acceptable random variation when a state of control exists. Generally speaking, the upper and lower control limits are set at $\pm$ three standard deviations from the mean. If normal probability distribution is assumed, these control limits will include 99.7 % of the random variations observed.

Process control chart



After a process has been brought to steady state operation, periodic samples are taken and plotted on the control chart. When the measurement falls within the control limits, the process is continued. If the measurement falls outside the control limits, the process is stopped and a search is made for an assignable causes. Through this procedure, the process is maintained in a constant state of statistical control and there is only natural variation in the processes output.

Quality measures: to detect abnormal variation, inspectors must be able to measure quality characteristics. Quality can be evaluated in two ways. One way is to measure **variables** i.e. product or service characteristics such as weight, length, volume or time that can be measured. Another way to evaluate quality is to measure **attributes**- i.e. product or service characteristics that can be quickly counted for acceptable quality. Generally, quality can be measured for control charts by *attributes* or by *variables*.

a. Process control with attribute measures: using P charts

The P chart is commonly used control chart for attributes. The quality characteristic is counted rather than measured and the entire item or service can be declared good or defective. For example in the bank industry, the attributed counted might be the number of incorrect financial statements sent. The method involves selecting a random sample, inspecting each item in it, and calculating the sample proportion defective -*which is the number of defective units (p) divided by the sample size (n)*. Sampling for P- charts involves yes- no decisions: the item or service either ‘is’ or “is not’ defective. The underling statistical distribution is based on the binomial distribution. However, for large sample size the normal distribution provides a good approximation to it.

To get the center line and control limits of the ‘P’ control chart, we take a large number of samples of ‘n’ units each. The P value is computed for each sample and then averaged over all samples to yield a value – $\bar{p}$. This value of $\bar{p}$ is used as the centre line, since it represents the best available estimate of the true average percent defective. We also use the value of $\bar{p}$ to compute upper and lower control limits. To construct the P chart, calculate:

$\bar{P} = \text{the sum of all sample} \div \text{no. of sample.}$

$$UCL = \bar{p} + 3\sqrt{((\bar{P} (1 - \bar{P})))/(\ n)}$$

$$LCL = \bar{p} - 3\sqrt{((\bar{P} (1 - \bar{P})))/(\ n)}$$

In this case, the process's standard deviation is the quantity under the square root sign. We are adding and subtracting three standard deviations from the mean to get the control limits. After the p control chart is constructed with this center line and lower control limits, samples of the process being controlled are taken and plotted on the chart i.e. the observed values of 'p' are plotted on the chart, one for each sample. If the sample percentage falls within the control limits, no action is taken. If the sample percentage falls outside the control limits, the process is stopped and a search for an assignable cause (material, operator, or machine) is made. After the assignable cause is found and corrected – or, in a very rare case, no assignable cause is found-the process is restored to operating condition and production is resumed.

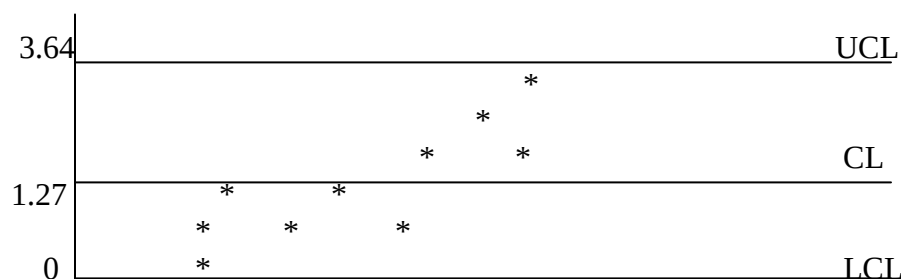
Example: suppose samples of 200 cards are taken from a key punch operation at 2 hours intervals to control the keypunch process. The percentage of cards in error for the past 10 samples is found to be 0.7, 1.2, 1.6, 2.0, 1.0, 0.8, 1.8, 1.5, 0.9, and 1.2 percent. Is the process out of control?

The average of these sample percentages yields a $\bar{p}=1.27$ percent or 0.0127 (sum of all samples divided by sample number i.e.10) which is the centre line of the control chart. The upper and lower control limits are:

$$UCL = 0.0127 + 3 \sqrt{\frac{0.0127(1-0.0127)}{200}} = 0.0364$$

$$LCL = 0.0127 - 3 \sqrt{\frac{0.0127(1 - 0.0127)}{200}} = -0.0110$$

When the LCL is negative, it is rounded up to 0 because a negative percentage is impossible. Thus we have the following charts.



Since all sample points are found to be in the control, these 10 samples can be used to establish the centre line and control limits.

Example:-The operations manager of the booking services department ABC bank is concerned about the number of wrong customer account numbers recorded by the ABC bank's personnel. Each week a random sample of 2500 deposit is taken, and the number of incorrect account numbers is recorded. The results for the past 12 weeks are shown in the following table. Is the process out of control?

Sample No.	Wrong account No.	Sample No.	Wrong account No.
1	15	7	24
2	12	8	7
3	19	9	10
4	2	10	17
5	19	11	15
6	4	12	3
Total			147